POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

DEVELOPMENT OF AN EMBEDDED SYSTEM ARCHITECTURE FOR SEMI-AUTOMATED PAPER-CHARCOAL BRIQUETTE MACHINE

A Thesis
Presented to the Faculty of Computer Engineering
Polytechnic University of the Philippines
Sta.Mesa, Manila

In Partial Fulfilment of the Requirements for the Degree
Bachelor of Science in Computer Engineering

by

Gero B. Añonuevo
Gerald R. Berongoy
Marco Gabriel P. Galdiano
Jester Carl C. Tiedra

May 2026

TABLE OF CONTENTS

Page

Title Page ..... **I**
Table of Contents ..... **II**
List of Tables ..... **III**
List of Figures ..... **IV**
Chapter 1: The Problem and its Setting **1**
Introduction 1
Theoretical Framework 3
Conceptual Framework 4
Statement of the Problem 7
Assumptions of the Study 8
Scope and Limitations of the Study 8
Significance of the Study 9
Definition of Terms 12
Chapter 2: Review of Literature and Studies **15**
Thematic Organization of Literature and Studies 15
Synthesis of the Reviewed Literature and Studies 34
Chapter 3: Methodology **36**
Research Design 36
Flowchart of Research Design 38
Description of Research Instrument Used 39
Mix Design 41
Data Gathering Procedure 42
Material Requirements 44
Software Requirements 54
Statistical Treatment 57
Design Project Flow 60
- Block Diagram 61
- Process Flowchart 62
- System Architecture 69
- Prototype Design 71

Specimen Details 75
Multiple Constraints 75

List of Tables

Table no.	Table description	Page
1	Mix design	41
2	Central Processing, Network Routing, and Memory Units	44
3	Autonomous Off-Grid Photovoltaic Power Infrastructure	45
4	Electromechanical Actuation, Fluidics, and Curing Modules	47
5	Closed-Loop Sensory Feedback and Fail-Safe Interlocks	50
6	Software Requirements	54
7	Environmental and Power Constraints	76
8	Material and Output Constraints	76
9	Process and Mechanical Constraints	77
10	Hardware and Sensor Constraints	78
11	Embedded System and Interface Constraints	79
12	Software and Network Constraints	79
13	Human-Machine Operational Constraints	80

List of Figures

Figure no.	Figure description	Page
1	Research Paradigm	5
2	Flowchart of Research Design/Process Flowchart	38
3	Block Diagram	61
4	Process Flow: Initialization and Diagnostics	63
5	Process Flow: Grinding and Soaking State	64
6	Process Flow: Mixing State	65
7	Flowchart: Compression and Molding Stage	67
8	Flowchart: Drying Stage	68
9	System Architecture of the Semi-Automated Paper-Charcoal Briquette Machine	69
10	Isometric View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine	71
11	Front View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine	72
12	Rear View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine	72
13	Left and Right Side Views of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine	73
14	Top View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine	73
15	Internal Mechanisms and Layout View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine	74
16	Expected Output Briquette Size of the Prototype	74

Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Due to rising Liquefied Petroleum Gas (LPG) prices, repurposing paper waste and residual charcoal fines into paper-charcoal briquettes is increasingly relevant. Philippine energy expenditure studies confirm that solid fuels are economic substitutes, leading households to shift demand to alternatives like charcoal as commercial energy costs rise (Dumagan & Abrigo, 2022). Utilizing these materials provides a practical method for localized waste reduction and an alternative solid fuel source, supported by local government initiatives promoting eco-friendly briquettes (Baccay, 2026). However, the production of this reliable heat source is significantly bottlenecked by its conventional reliance on manual labor.

The manual process for producing extended-burn paper-charcoal briquettes is inefficient, time-consuming, and physically taxing. The workflow demands constant human labor during grinding, soaking, mixing, and molding, compounded by lengthy sun-drying periods. This reliance on manual effort severely limits production throughput and increases man-hour consumption. Furthermore, human fatigue and manual pressing result in critical inconsistencies across briquette size, compaction force, moisture content, and final calorific quality.

This operational bottleneck is confirmed by briquette-making research, which identifies the inadequacy of production technology, rather than slurry chemistry, as the

main barrier to accessibility (Mwamlima et al., 2023). The current industry employs two extremes: capital-intensive, fully automated industrial plants that require three-phase grid power, making them unfeasible for off-grid community deployment; or readily available, cheap manual presses. Manual presses, however, fail to provide controlled and repeatable operating parameters, resulting in unpredictable solid fuel density (Cabaces et al., 2023).

To bridge the gap between manual inconsistency and industrial-scale output, purely mechanical presses must be transformed into Computer Engineering-rooted cyber-physical systems. Integrating an embedded architecture replaces reliance on human intuition for moisture ratios and compaction pressure with deterministic, sensor-driven logic. Advancements demonstrate this efficacy: utilizing ADCs, Hall effect sensors, and servo-driven actuators enables automated, precise classification of briquette moisture content via electrical resistance (Pniel, 2026). Governing these stages with IoT-enabled microcontrollers (e.g., ESP architecture), load cells, and HX711 amplifiers ensures exact compaction limits and weight consistency (Thongsan et al., 2024). This computational approach guarantees precision and structural repeatability for localized automation, facilitating real-time telemetry to PWA dashboards and HMIs to improve administrative monitoring and operational efficiency.

This study aims to develop a solar-powered, semi-automated paper-charcoal briquetting machine to streamline production. The system utilizes a computerized control architecture to coordinate the main production stages through a semi-automated, human-in-the-loop control architecture, including grinding, soaking, mixing, compression, and drying. Through a human-in-the-loop framework, an

operator manages the workflow via an on-board touchscreen or a wireless dashboard. This design optimizes off-grid solid fuel production, ensuring consistent briquette quality while significantly reducing manual labor and dependence on the electrical grid.

Theoretical Framework

This study is theoretically anchored on the Agile Hardware Development Framework. In the development of a complex cyber-physical system, traditional linear (Waterfall) engineering models are highly prone to hardware-software mismatch. To mitigate this risk, the machine's architecture was developed using an Agile Hardware approach. This framework dictates that physical mechanics, sensor networks, and software logic must be co-designed through continuous, testable iterations rather than a single final assembly.

In this study, the Agile framework provides the methodological justification for the iterative integration of the ESP32. Instead of writing the complete firmware blindly, the Finite State Machine (FSM) was developed and tested in isolated stages alongside the mechanical subsystems such as the 775 motors for pulverization and the BTS7960 linear actuator for compression.

This iterative "sprint" methodology allowed for immediate hardware fault isolation. Specifically, it enabled the continuous testing and refinement of the power isolation strategy, ensuring that the back-EMF from the inductive bus was successfully isolated from the logic core before the system advanced to evaluative testing.

By adopting Agile as the theoretical lens, this research shifts the engineering process from a rigid assembly task to a responsive, fault-tolerant evolution. It ensures that the cyber-physical architecture is strictly validated for mechanical stability, electrical safety, and logic accuracy prior to continuous operational deployment.

Conceptual Framework

The research paradigm illustrated in Figure 1 was utilized to guide the development of an embedded system architecture for a self-sustaining, solar-powered, and semi-automated paper-charcoal briquette machine. The Input-Process-Output (IPO) model was specifically adopted to provide a structured conceptualization of the entire research and development workflow.

The foundational inputs include Raw Materials (physical ingredients), Hardware Components (microcontrollers, sensors, and mechanical modules), and Software Components (digital interfaces and logic frameworks). These flow through the process block's developmental phases to achieve the main objective: "Development of an Embedded System Architecture for Semi-Automated Paper-Charcoal Briquette Machine." A feedback block establishes a closed-loop system for continuous system refinement.

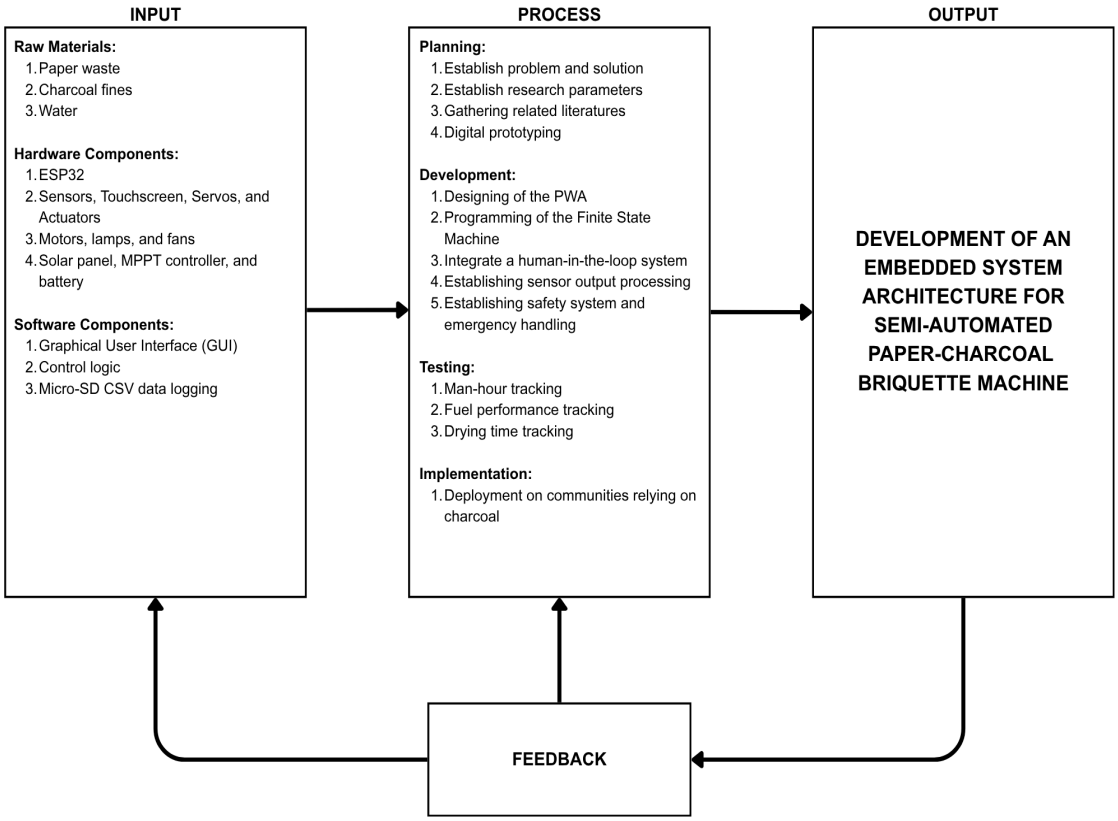


Figure 1. Research Paradigm

The inputs required for the system's physical and technical operation are divided into three main categories. First, the Raw Materials consist of paper waste, charcoal fines, and water. Second, the Hardware Components required for the system to function effectively include the ESP32 microcontroller, sensors, a touchscreen, servos, actuators, motors, lamps, fans, alongside an autonomous power infrastructure comprising a solar panel, MPPT controller, and battery. Third, the Software Components utilized in the system's development comprise the Graphical User

Interface (GUI), the core control logic, and Micro-SD CSV data logging to ensure efficient data tracking and machine operation.

The processes involved in the research consist of a structured methodology used to achieve the study's objectives. It begins with Planning, which involves establishing the problem and solution, defining research parameters, gathering related literature, and creating digital prototypes. This transitions into the Development phase, focusing on designing the Progressive Web App (PWA), programming the Finite State Machine (FSM), integrating a human-in-the-loop system, establishing sensor output processing, and configuring the safety system and emergency handling protocols. Following development is the Testing phase, which employs methods for man-hour tracking, fuel performance tracking, and drying time tracking to validate system efficiency. Finally, the Implementation phase involves the deployment of the system in communities relying on charcoal.

These inputs and processes produce the final outcome of this study. The feedback loop continuously evaluates the testing metrics and implementation results, routing them back into the input and process stages to guarantee the optimized development of the semi-automated paper-charcoal briquette machine.

Statement of the Problem

This study aims to develop a system that innovates the production of paper-charcoal briquettes, integrating grinding, soaking, mixing, molding, and drying, to create a high-efficiency fuel source with minimal human intervention. Specifically, the study seeks to answer the following:

1. What are the prevailing challenges and inefficiencies in the utilization of paper waste and charcoal fines as an alternative solid fuel source?
2. What are the differences between the semi-automated system and the traditional manual briquette-making process in terms of physical demands, time consumption, and continuous operational oversight?
3. What are the design and operational limitations of the proposed semi-automated system, particularly concerning manual material feeding, water-insoluble toxin extraction, and standardized structural boundaries?
4. What is the level of effectiveness of the developed semi-automated machine in improving production efficiency and product quality, specifically measured by:
    - 4.1. Total man-hour consumption
    - 4.2. Throughput time
    - 4.3. Production rate
    - 4.4. Fuel performance

5. What is the level of acceptability of the automated system, including its Graphical User Interface (GUI), based on ISO 25010 quality model standard, when evaluated?

Assumptions of the Study

This developmental research is anchored on the following basic assumptions:

- Operational Assumption: Mechanizing the soaking, mixing, and molding phases of briquette production will significantly increase the production rate and ensure a more uniform briquette density compared to traditional manual methods.
- Material Assumption: The natural cellulose fibers present in soaked waste paper possess sufficient adhesive properties to effectively bind residual charcoal fines without the need for synthetic chemical additives.
- Economic Assumption: There is a sustained local demand for affordable alternative solid fuels due to the escalating and volatile prices of commercial Liquefied Petroleum Gas (LPG).
- System Viability Assumption: A semi-automated pipeline provides a more practical, cost-effective, and maintainable solution for small-scale solid fuel production compared to fully automated industrial machinery.

Scope and Limitations of the Study

The study focuses on developing an embedded architecture for a solar-powered machine that produces rectangular paper-charcoal briquettes optimized for prolonged burn times. The system automates the processing of water,

charcoal fines, and waste paper through sequential electromechanical stages: grinding, soaking, mixing, molding, and halogen-assisted drying. Operations are managed via a dual-interface control framework. An on-machine touchscreen serves as the primary hub for step-by-step process authorization using audio prompts, while a localized progressive web app dashboard provides real-time telemetry monitoring and streamlined administrative data management. The system will be powered by a solar panel system for self sustenance.

The project is bound by specific mechanical, material, and operational constraints. Briquette output is restricted to a single fixed dimension, and the lack of a pre-soaking purification mechanism means insoluble impurities in the waste paper are retained in the final product. Furthermore, the workflow is strictly semi-automated. Human intervention is required to manually feed raw materials, physically transfer molded briquettes to the drying racks, and authorize each process phase via the touchscreen interface, thereby precluding fully autonomous production.

Significance of the Study

The development of the semi-automated paper-charcoal briquette machine offers a multifaceted impact that addresses both immediate socio-economic needs and long-term environmental sustainability. The significance of this study is categorized into its contributions to global goals, national and local sectors, and academic advancement.

United Nations' Sustainable Development Goals (SDGs)

The proposed system directly aligns with several key UN SDGs, specifically Goal 7 (Affordable and Clean Energy), Goal 12 (Responsible Consumption and Production), and Goal 13 (Climate Action). By repurposing discarded paper waste and residual charcoal fines into an efficient solid fuel, the machine promotes a circular economy and reduces dependence on expensive, non-renewable commercial fuels among low-income communities. Furthermore, its integration of a solar panel as a self-sustaining power source directly reinforces Goal 7 by ensuring that productive energy technology remains accessible even in off-grid or electricity-limited settings. This contributes to a broader, sustainable approach to energy consumption and waste management at the community level.

Government and Waste Management Sectors

For the Municipality of Cavinti, Laguna, and its local government unit, the system presents a practical, community-scale solution to existing waste management challenges. By intercepting paper and biomass waste before it reaches landfills, the machine supports the municipality's local waste diversion efforts and reduces the burden on existing disposal infrastructure. Furthermore, it supports governmental environmental initiatives by providing a viable alternative to wood charcoal, thereby indirectly helping to curb the demand that drives illegal tree cutting and timber poaching, which the Department of Environment and Natural Resources (DENR) has directly linked to the unaffordability of commercial cooking fuels nationwide.

Households and Local Communities

The primary and most direct beneficiaries of this system are the households and communities of Cavinti, Laguna, particularly those who rely on solid fuels such as charcoal and wood coal as their primary means of cooking due to limited or no access to Liquefied Petroleum Gas (LPG) and electricity. For these community members, commercially purchased charcoal is not merely an alternative but often their only available and affordable heat source. However, the cost and physical effort associated with sourcing charcoal remain persistent burdens. By automating the labor-intensive stages of briquette production and powering the system through a self-sustaining solar panel, the machine empowers local residents and small-scale fuel producers in Cavinti to manufacture high-quality, extended-burn briquettes with reduced physical effort and at minimal recurring cost, providing a reliable, community-adoptable cooking fuel that does not depend on commercial energy infrastructure.

Solid Fuel Manufacturers and Small Businesses

For small enterprises in and around Cavinti, Laguna that rely on the production or resale of alternative solid fuels, the semi-automated pipeline resolves the critical challenges of inefficient production rates and inconsistent product quality that characterize manual briquette-making. The integration of a Graphical User Interface (GUI) ensures intuitive operation that can be readily adopted by workers without specialized technical training, allowing local businesses to scale their production of uniform, high-density briquettes more effectively and at greater volume.

Future Researchers

The documentation, baseline metrics, and system architecture established in this research will serve as a foundational reference for future academic endeavors. Future researchers can utilize this study to benchmark efficiency, innovate upon the semi-automated pipeline, or explore further integration of smart technologies in waste-to-energy systems.

Definition of Terms

For a clearer understanding of this study, the following terms are defined operationally and conceptually as they are used in the context of the research:

CHARCOAL FINES - Residual, leftover charcoal fragments and dust that serve as one of the primary raw materials to be processed and repurposed by the semi-automated machine.

CYBER PHYSICAL SYSTEMS - Systems that involve simultaneous development of both hardware and software. In which the two parts need seamless coordination for successful operation.

PAPER-CHARCOAL BRIQUETTES - The final solid fuel product produced by the machine, engineered from a paper-charcoal mixture to serve as a reliable heat source that burns longer than standard, store-bought charcoal.

GRAPHICAL USER INTERFACE - Refers to the visual or graphical aspect of the system that serves as the medium of communication between the users and the machine.

HUMAN-IN-THE-LOOP - A design framework requiring a human operator to monitor alerts and manually authorize transitions between automated stages via the GUI.

LITTLEFS - the internal filesystem partition used by the microcontroller to host the static React Progressive Web App (PWA) web assets.

PROGRESSIVE WEB APP - A web app that behaves like a native app, including offline support.

MAN-HOUR CONSUMPTION - A metric used in this study's Time and Motion evaluation to quantify the total human labor time required for production, which the semi-automated system aims to significantly reduce.

PAPER-CHARCOAL BRIQUETTE - An alternative or supplementary solid chunk fuel created by the automated machine through the combination, compression, and drying of water, paper waste, and charcoal dust.

PRODUCTION RATE - A quantitative measure of the automated system's efficiency, calculated as the number of briquette units successfully manufactured per hour (units/hr).

SEMI-AUTOMATED PIPELINE - Refers to the integrated mechanical system developed in this study that performs the sequential stages of grinding, soaking, combining, molding and compression, and drying with minimal human intervention.

THROUGHPUT TIME - The total duration required for raw materials to pass through the entire semi-automated production cycle, from the initial manual feeding and grinding to the final drying phase, yielding a ready-to-use briquette.

WATER RETENTION VALUE (WRV) - The ability of the kraft paper to hold water in the scenario of being soaked then mechanically compressed to decrease the water content.

WATER HOLDING CAPACITY (WHC) - The ability of charcoal fines to hold water after being soaked from the water of the soaked kraft paper then being compressed mechanically.

Chapter 2

REVIEW OF LITERATURE AND STUDIES

Thematic Organization of Literature and Studies

The increasing cost of commercial household energy sources continues to pressure Filipino households to seek cheaper and more accessible alternatives. As electricity, LPG, and other commercial energy sources become difficult to sustain for low-income and rural users, many households continue to rely on traditional solid fuels such as wood and charcoal (Dumagan & Abrigo, 2022; Pantuhan et al., 2024). However, this dependence creates environmental concerns, especially when rising charcoal demand contributes to illegal tree cutting, timber poaching, and pressure on local forest resources (Meniano, 2026). At the same time, communities continue to face problems in managing paper waste and charcoal fines that could potentially be redirected into useful fuel products (Anonas et al., 2023; Baccay, 2026).

Paper-charcoal briquettes present a practical response to these connected problems because they combine waste reduction with alternative fuel production. Paper waste can serve as a natural binder, while charcoal fines can be densified into a usable solid fuel when properly mixed, molded, compressed, and dried (Chipangura et al., 2024; Lomunyak et al., 2024). However, manual briquette production remains limited by physical labor, inconsistent material preparation, variable molding pressure, long processing time, and the need for continuous human supervision. These

limitations justify the development of a semi-automated system that integrates grinding, soaking, mixing, molding, and drying into a more controlled production process.

This chapter reviews literature and studies related to the development of an embedded system architecture for a semi-automated paper-charcoal briquette machine. The discussion covers the challenges in using paper waste and charcoal fines, the differences between automated and manual briquette production, the design and operational limitations of the proposed system, the expected effectiveness of the machine in production efficiency and product quality, and the acceptability of the embedded system and graphical user interface based on ISO/IEC 25010.

1. Challenges and Inefficiencies in Utilizing Paper Waste and Charcoal Fines

The need for alternative solid fuel is strongly connected to the economic pressure experienced by households that rely on commercial energy. In the Philippines, household energy demand is sensitive to price changes, and solid fuels remain substitutes when electricity, gas, and liquid fuels become financially difficult to sustain (Dumagan & Abrigo, 2022). Castillo-Dida (2024) further links electricity pricing and carbon emissions, showing that energy decisions affect both household cost and environmental outcomes. Pantuhan et al. (2024) add that rural communities in a southern Philippine city often limit the use of electricity and LPG for cooking because of cost, hazards, and access concerns, making conventional wood and charcoal fuels persistent in daily life.

The same pattern appears in international literature on domestic fuel use. Kalli et al. (2022) found that even subsidized LPG programs do not automatically eliminate

traditional fuel use because households continue to consider refill cost, income level, and availability. Habib et al. (2023) similarly describe fuel stacking, where households use LPG together with solid biomass fuels instead of completely shifting to cleaner fuel. Laska and Ige (2023) explain that in settings where electricity supply is insufficient or expensive, firewood and charcoal remain dominant domestic fuel sources. These studies support the first SOP by showing that solid fuel dependence is not merely a cultural preference but an economic and access-driven challenge.

Local reports also show that the renewed demand for charcoal creates environmental pressure. Magbanua (2026) reported a decline in LPG use as households shifted toward firewood during periods of high fuel cost, while Meniano (2026) connected the rising demand for wood charcoal with illegal tree cutting and timber poaching. These conditions make paper-charcoal briquettes relevant because they can reduce dependence on freshly cut wood charcoal while using materials that are otherwise treated as waste.

The waste side of the problem is equally important. Anonas et al. (2023) describe the Philippines as facing growing solid waste generation and limited disposal infrastructure, creating a policy push toward waste-to-energy solutions. Burigsay (2025) found that community-level solid waste management enterprises, including charcoal briquetting, can be feasible when supported by capacity building and basic equipment. Baccay (2026) also documented a DENR initiative in Batanes that converted paper waste into eco-friendly charcoal briquettes, showing that paper-based briquetting is already recognized as a practical local waste-reduction strategy.

Paper waste is not automatically an ideal fuel, however, because its performance depends on paper type, fiber quality, binder behavior, and contamination. Felongco et al. (2025) found that cardboard-based paper briquettes had better compressive strength, longer burn duration, and lower porosity than office paper and newspaper briquettes, showing that paper waste must be sorted and selected carefully. Genuino et al. (2022) also found that paper pulp can function as a binder in bio-briquettes, but the best performance depends on the base material and binder combination. These findings show that paper waste utilization involves material variability that must be addressed before efficient production can occur.

Charcoal fines present a related inefficiency. Chipangura et al. (2024) describe charcoal fines as a useful but underutilized residue that requires binding and densification to become a usable solid fuel. Lomunyak et al. (2024) show that loose charcoal dust can be converted into briquettes using paper, starch, or algae binders, but the resulting density, shatter resistance, calorific behavior, and emissions vary by binder proportion. Waheed et al. (2022) further explain that powdered or loose biomass creates handling, transportation, storage, and air-pollution problems unless it is densified into higher-density briquettes. Thus, the main challenge is not only the availability of waste paper and charcoal fines but the inefficient condition of these materials before grinding, soaking, mixing, molding, and drying.

2. Differences Between Automated and Traditional Manual Briquette Production

Traditional briquette-making is commonly labor-intensive because it requires repeated manual preparation, mixing, molding, pressing, and drying. wondimagegn and Edao (2023) describe conventional briquette preparation steps in which waste paper is used as a binder, water assists mixing, and materials are manually combined before pressing. Lomunyak et al. (2024) produced charcoal-dust briquettes using a manually operated machine and showed that binder proportion affected density, shatter resistance, calorific value, burning rate, and emissions. These studies prove that manual methods can produce usable briquettes, but they also show that manual production depends heavily on human consistency in mixing, proportioning, and pressure application.

The physical demand of manual production is reduced when the process is transferred to machine-assisted equipment. Tesfaye et al. (2023) developed and evaluated a coffee husk biomass briquette machine and measured throughput capacity, densification, resistance to water penetration, shatter resistance, tumbling resistance, calorific value, fixed carbon, and ash content. wondimagegn and Edao (2023) also evaluated a pellet press briquette machine and reported machine capacity and efficiency as key indicators. These studies show that machine-based briquetting changes the work from repetitive physical compression to machine setup, feeding, monitoring, and output handling.

Automated or semi-automated systems also differ from manual production in time consumption. Cabaces et al. (2023) developed a low-cost briquetting system with defined mixing speed, mixing time, compressive force, and briquetting time, producing

twenty briquettes under tested operating conditions. This kind of controlled operation contrasts with traditional manual methods because the time for each phase can be measured, repeated, and improved. Escalante and Alcaraz (2025) further showed that auger speed significantly affected machine output capacity, energy consumption, kindling time, briquetting efficiency, density, moisture resistance, shatter resistance, and burn time in an automated screw-type charcoal briquetting machine.

Lean manufacturing literature explains why this mechanical difference matters. Alora and Amaya (2025) found that motion, transportation, and overprocessing extended cycle time and reduced output in a manufacturing process, while lean improvements significantly reduced cycle time and increased productivity. In the proposed briquette machine, unnecessary walking, repeated hand-mixing, manual pressing, and constant visual checking are comparable non-value-added activities. Therefore, automation is not only a mechanical upgrade but also a workflow redesign that reduces wasted movement and waiting time.

Time and Motion Study supports the comparison between manual and automated production. Jesus et al. (2025) developed a data-driven time and motion study tool that reduced analysis effort and processing time by approximately six times while enabling more detailed recording of workstation events. This supports the proposed study's need to compare total man-hour consumption, throughput time, and production rate between the traditional method and the automated system. It also justifies using measured time data instead of relying only on observation or user perception.

The difference in continuous operational oversight is also central. Manual briquetting requires the operator to remain involved in nearly every production step, while semi-automation can shift the operator's role toward loading materials, selecting system settings, monitoring status, and removing finished briquettes. Escalante and Alcaraz (2025), Cabaces et al. (2023), and Tesfaye et al. (2023) all demonstrate that machine performance can be evaluated through controlled variables rather than pure manual judgment. This directly supports the second SOP because the proposed system is expected to reduce physical demands, shorten production time, and reduce continuous hands-on supervision.

3. Design and Operational Limitations of the Proposed Semi-Automated System

The proposed system still has design and operational limitations even if it reduces manual labor. One limitation is manual material feeding. Ibitoye et al. (2022) found that compaction pressure, blending ratio, and particle size substantially affected the combustion and physico-mechanical characteristics of composite fuel briquettes, while Yusuf et al. (2023) found that compaction pressure changed density, stability, durability, combustion temperature, and combustion rate in rice husk charcoal briquettes. Cabaces et al. (2023) also established fixed operating conditions for a briquetting system, including mixing speed, compressive force, and briquetting speed, to produce standardized briquettes. These findings indicate that briquette quality is sensitive to material proportion, particle condition, and compaction settings. Therefore, for the proposed system, controlled manual feeding is treated as an operational requirement to help maintain consistent input conditions and reduce the risk of uneven mixing or excessive machine load.

Feedstock variability also limits the grinding and mixing stages. Tesfaye et al. (2023) found that binder ratio affected throughput capacity, densification, water resistance, shatter resistance, ash content, fixed carbon, and calorific value. Lomunyak et al. (2024) similarly found that paper, starch, and algae binders produced different density ranges, shatter resistance, and emission levels. These findings show that the machine cannot guarantee the same output quality unless material ratios and moisture levels are standardized before operation.

Another limitation concerns contaminants in paper waste. Mofokeng et al. (2024) found that recycled paper grades may contain chemical compounds associated with additives, plastics, consumer exposure, sorting, and reprocessing, and that different separation approaches may be needed to reduce contaminant exposure. Mentes et al. (2024) also found that burning paper-based waste with wood can alter emitted particles and increase polycyclic aromatic hydrocarbon concentrations depending on the waste type. These studies support the limitation stated in the SOP, but they should not be interpreted as proof that water soaking removes toxins. Instead, soaking should be treated only as a fiber-softening and pulping step, not as a contaminant-removal process, since paper-based waste may still contain inks, coatings, additives, plastics, or chemical residues that require material selection and separation controls.

The literature on paper fibers also shows that water handling is technically important but not a complete purification method. Sjostrand and Brolinson (2022) found that additives such as polyvinylamine can affect pulp drainage, dewatering, tensile strength, and air permeance, showing that paper fiber behavior depends on

chemistry, water retention, and dewatering conditions. Arabi and Dahmardeh Ghalehno (2023) likewise show that drying lignocellulosic biomass involves simultaneous heat and mass transfer and that moisture removal is nonlinear. Therefore, the proposed system must treat soaking and drying as controlled preparation steps rather than as complete detoxification or uniform quality assurance.

The machine also has standardized structural boundaries. Cabaces et al. (2023) produced briquettes with fixed dimensions under a specific molder and compression setup, while Escalante and Alcaraz (2025) showed that auger speed affected output quality in a screw-type briquetting machine. Yusuf et al. (2023) found that compaction pressure affects density, stability, durability, combustion temperature, and combustion rate in rice husk charcoal briquettes. Ibitoye et al. (2022) similarly found that compaction pressure, blending ratio, and particle size substantially affected the combustion and physico-mechanical properties of composite fuel briquettes. These findings justify limiting the proposed machine to a defined briquette size, shape, mold volume, and actuator force range.

The drying stage also has operational limits. Tuan et al. (2023) found that material thickness and drying temperature significantly affect drying time and equilibrium moisture content under a halogen dryer. Alabi et al. (2023) found that moisture content, density, size, and temperature profiles significantly influence briquette combustion behavior, mass loss, and burnout time. These studies mean that a halogen-assisted dryer must be designed around a fixed briquette thickness and expected moisture range. If the briquette is too thick, too wet, or unevenly compacted, drying time and final fuel performance may vary.

Power supply is another boundary of the proposed system. Alfanz et al. (2023) showed that solar power can support smart agricultural equipment, while Nkuriyingoma et al. (2022) demonstrated that photovoltaic systems with battery storage can support small household loads in off-grid conditions. However, these sources also imply that solar-powered operation depends on panel sizing, battery capacity, load demand, and solar availability. Thus, the proposed system may be self-sustaining, but its operating schedule and motor load must remain within the limits of the solar-battery design.

4. Effectiveness of the Developed Machine in Production Efficiency and Product Quality

Effectiveness must first be measured through labor and time savings. Alora and Amaya (2025) show that cycle time and output can be used to identify production inefficiencies, especially those caused by unnecessary motion, transportation, manual handling, and redundant steps. Jesus et al. (2025) show that time and motion study can be strengthened through data-driven recording of workstation events. These studies support the use of total man-hour consumption and throughput time as effectiveness indicators for the proposed machine.

Production rate is another major measure. Cabaces et al. (2023) evaluated a briquetting system using mixing rate, briquetting rate, percent yield, and energy efficiency. Escalante and Alcaraz (2025) reported that an automated screw-type charcoal briquetting machine reached its highest output capacity at a specific auger speed range, while another speed range produced higher briquetting efficiency and stronger briquettes. Tesfaye et al. (2023) and wondimagegn and Edao (2023) also measured machine capacity in evaluating briquetting equipment. These studies show

that production rate should be measured together with quality because the fastest machine setting is not always the best setting for briquette strength or fuel performance.

Fuel performance depends heavily on density, moisture, and compaction. Yusuf et al. (2023) found that compacting pressure affects the mechanical and combustion characteristics of rice husk charcoal briquettes. Waluyo et al. (2022) found that blending ratio affected stable density, water resistance, compressive strength, and higher heating value in sawdust-cocoa pod briquettes. Ibitoye et al. (2022) found that compaction pressure, particle size, and blending ratio affected calorific value, density, shatter index, and water resistance. These findings support measuring product quality beyond visual appearance.

Binder performance is also a key indicator of output quality. Obi et al. (2022) reviewed biomass briquette binders and emphasized that binders improve cohesive strength and affect mechanical and handling properties. Malovanyy et al. (2023) found that adding a natural binder could reduce required forming pressure, reduce engine power, and increase briquette density. Montiel et al. (2023) found that algal biomass can function as a binder for eco-briquettes, while Lomunyak et al. (2024) found that paper-bound charcoal dust briquettes achieved high shatter resistance. These studies support using soaked paper pulp as a natural binder while still testing whether the final product is durable.

Burning performance must also be tested through combustion indicators. Felongco et al. (2025) found that cardboard-based paper briquettes produced higher compressive strength, longer burn duration, and lower porosity than other paper types.

Dumrique et al. (2024) tested agricultural residue briquettes through water-boiling, temperature, burning rate, and acceptability measures, showing that fuel performance can be evaluated through practical cooking-related indicators. Waheed et al. (2022) explain that densification improves handling and energy density while addressing the limitations of loose biomass. These sources support the study's use of fuel performance as one of the main effectiveness measures.

Moisture control is directly tied to both throughput and combustion quality. Arabi and Dahmardeh Ghalehno (2023) found that halogen drying temperature affected drying time and drying rate in lignocellulosic biomass, while Tuan et al. (2023) found that material thickness and temperature influenced moisture content prediction under halogen drying. Alabi et al. (2023) further showed that moisture content and temperature profiles influence briquette combustion and burnout. Therefore, effectiveness must include not only how many briquettes are produced but whether the drying system produces briquettes with predictable moisture levels.

Environmental and health-related fuel performance also matters. Qi et al. (2022) found that improved biomass combustion technology can reduce pollutant emissions, including large reductions in PM2.5 under improved stove conditions. Glalah et al. (2024) found that binder type affected CO and PM2.5 emissions from biochar briquettes and reported several briquette formulations within safe air-quality thresholds for domestic use. Mentes et al. (2024), however, warn that burning mixed waste materials can increase harmful PAH concentrations. These findings show why the developed machine must produce consistent, well-dried briquettes from selected feedstock rather than uncontrolled mixed waste.

Finally, effectiveness includes the machine's ability to operate in the intended setting. Alfanz et al. (2023) demonstrate the feasibility of applying solar power to smart agricultural systems, and Nkuriyingoma et al. (2022) show that photovoltaic systems with battery storage can provide reliable off-grid power for small household applications. For the proposed machine, this supports evaluating whether the solar-powered embedded system can sustain grinding, soaking support, mixing, molding, and drying operations without depending entirely on grid electricity.

5. Acceptability of the Semi-Automated System and GUI Based on ISO 25010

Acceptability should be understood as both technical acceptability and user acceptability. Burigsay (2025) found that community-based solid waste management enterprises, including charcoal briquetting, are viable when they are supported by equipment, capacity building, and stakeholder participation. Mwamlima et al. (2023) also note that briquette-making technologies can support sustainable cooking energy but may be limited by complexity and cost. These findings suggest that the proposed machine must not only work technically; it must also be understandable, manageable, and acceptable to the intended users.

ISO/IEC 25010:2023 provides the appropriate framework for evaluating the embedded system and GUI because it defines a product quality model for ICT and software products and supports requirements specification, testing objectives, quality control criteria, acceptance criteria, and product-quality measures (International Organization for Standardization & International Electrotechnical Commission [ISO/IEC], 2023). Since the proposed machine includes an embedded controller and graphical user interface, ISO/IEC 25010 is suitable for evaluating whether the interface

supports the user's required tasks, gives understandable feedback, performs reliably, and assists safe operation.

Recent software-quality literature supports applying ISO/IEC 25010 to user-facing interfaces, although the context of each study must be considered. Carrión-León et al. (2025) evaluated the interaction capability of a serious-game graphical interface for children with autism spectrum disorder using ISO/IEC 25010:2023, focusing on operability and inclusiveness through interface interaction data such as clicks, visual focus, level selection, feedback components, time, moves, and scores. While this is not an industrial machine study, it is useful as an example of how ISO/IEC 25010 can be operationalized into observable GUI criteria. In the proposed machine, the transferable point is the evaluation approach: users must be able to select or monitor processes such as grinding, soaking, mixing, molding, drying, and system status without confusion.

Functional suitability is important because the GUI must cover the actual machine operations required by the study. The interface should provide controls and indicators for process activation, timer or cycle monitoring, drying status, motor status, and completion alerts. If any of these functions are missing or unclear, the user may still need continuous manual oversight, weakening the purpose of automation. Thus, functional suitability directly connects to the SOP's concern for minimal human intervention.

Usability or interaction capability is equally important because the system may be used by students, operators, or community users who are not specialists in embedded systems. The GUI should use clear labels, visible process states, simple

prompts, and error alerts to reduce operator confusion. Reliability and performance efficiency must also be evaluated because delays, freezes, unreadable values, or inconsistent button responses can interrupt production and reduce trust in the machine.

Safety and maintainability are also part of acceptability. Since the machine includes moving parts, heat, moisture, and solar-powered electrical components, the GUI must support safe operation through warnings, status feedback, and predictable controls. The solar-power studies of Alfanz et al. (2023) and Nkuriyingoma et al. (2022) support the feasibility of localized power systems, but ISO/IEC 25010 helps determine whether the user-facing control system remains acceptable during actual operation. Therefore, the acceptability evaluation should include both user ratings and observed system behavior based on selected ISO/IEC 25010 quality characteristics.

6. Production Process and System Workflow of Paper-Charcoal Briquette Making

The production process of paper-charcoal briquette making involves paper preparation, mixing, molding, compression, and drying. These stages explain how the proposed machine converts waste paper and charcoal fines into a usable briquette product.

Paper waste preparation serves as the initial stage of the production workflow. Genuino et al. (2022) show that paper pulp can function as a binder in bio-briquettes, while Felongco et al. (2025) found that different paper types produce different briquette outcomes in terms of compressive strength, porosity, and burning duration. Lomunyak et al. (2024) also demonstrated that paper can be used as a binder for charcoal-dust

briquettes, but binder proportion affects density, shatter resistance, calorific value, burning rate, and emissions. These findings support soaking and pulping as a process step, but only for fiber softening and binder preparation, not toxin extraction.

Mofokeng et al. (2024) found that recycled paper grades may contain chemical compounds associated with additives, plastics, consumer exposure, sorting, and reprocessing, while Mentes et al. (2024) showed that burning paper-based waste with wood can alter emitted particles and increase harmful compound concentrations depending on waste type. Therefore, soaking is treated only as a fiber-softening and pulping step, while paper sorting and contaminant avoidance remain design and operational limitations.

Mixing wet paper pulp with charcoal fines is the next stage of the production workflow. Chipangura et al. (2024) describe charcoal fines as an underutilized residue that requires binding and densification before it can become a usable solid fuel. Malovanyy et al. (2023) found that natural binders can improve briquette formation by reducing forming pressure and increasing briquette density, while Obi et al. (2022) explain that binders improve cohesive strength and handling properties in biomass briquettes. These studies support the system's mixing stage because it is intended to promote contact between wet paper fibers and charcoal particles before molding and compression.

Mechanical molding and compression form the next stage of the production workflow. Cabaces et al. (2023) developed a low-cost briquetting system with defined mixing speed, compressive force, briquetting speed, and fixed briquette dimensions, showing that machine settings can standardize output geometry. Yusuf et al. (2023)

found that compaction pressure affects briquette density, stability, durability, combustion temperature, and combustion rate, while Ibitoye et al. (2022) found that compaction pressure, blending ratio, and particle size affect combustion and physico-mechanical properties. These findings justify the molding and compression stage because it directly connects the mechanical design to the expected product quality.

Controlled drying is the final stage of the production workflow. Tuan et al. (2023) found that halogen drying temperature and material thickness influence drying time and predicted moisture content, while Arabi and Dahmardeh Ghalehno (2023) show that lignocellulosic biomass drying involves changing moisture movement and drying-rate behavior. Alabi et al. (2023) further connect moisture content, density, size, and internal temperature profiles to briquette combustion behavior and burnout time. These studies justify thermal drying as a final process because the briquette's fuel performance depends on moisture reduction, not only on successful molding.

7. Theoretical Framework: Agile Hardware Development for Cyber-Physical System Integration

This study is guided by an Agile hardware-software development framework. Drutchas and Eppinger (2022) explained that Agile began in software development but has increasingly been applied to combined hardware and software projects, where teams must adjust Agile practices because hardware work involves physical constraints, sprint-cadence limitations, and backlog-creation challenges. This makes Agile hardware development suitable as a guiding framework for the proposed paper-charcoal briquette machine because the system combines mechanical parts,

motors, a linear actuator, power electronics, embedded firmware, sensor feedback, and a graphical user interface.

The key value of Agile hardware development is that it treats system design as a sequence of testable iterations rather than one final assembly. Drutchas and Eppinger (2022) identify combined hardware-software development as a setting where coordination and integration points must be managed carefully. For this study, that means the grinder motor, mixing subsystem, molding actuator, drying element, power supply, ESP32 controller, and GUI should be developed and checked in stages. Each stage may be treated as an iterative subsystem validation step where mechanical behavior, electrical stability, and control logic are checked before the next subsystem is integrated.

Fagcang et al. (2022) support this staged approach through their review of component-in-the-loop and hardware-in-the-loop methods, which combine physical subsystems with real-time virtual or control environments to accelerate development and integration. This principle is applicable to the proposed machine because each physical subsystem can be tested under controlled conditions before full-system deployment. In the proposed machine, this supports isolating and testing the 775 motor for grinding, the mixing motor, the BTS7960-driven linear actuator for compression, and the drying control before evaluating continuous operation.

The embedded software side of the framework is supported by Finite State Machine logic. Barkalov et al. (2024) describe finite state machines as structured digital systems whose behavior depends on defined states, transitions, inputs, outputs, transition functions, and output functions. In the proposed system, this FSM logic can

be applied to firmware organization by separating operations into distinct states such as idle, grinding, soaking support, mixing, molding, drying, complete, and fault. Using an FSM-based firmware structure supports modular testing because each machine state can be checked against its expected inputs, outputs, and transitions before full system integration.

The Agile hardware framework also supports early electrical-risk isolation. Figueiroa and Torres (2022), in a microcontroller-based smart greenhouse design, described sensors and actuators controlled through electrical subsystems and discussed the inductive flyback effect in a fan circuit, using a flyback diode as the protection solution. A comparable concern applies to the proposed briquette machine's inductive loads, such as motors and actuator circuits, where protection circuits should be validated before continuous machine trials. Therefore, the logic controller, motor drivers, and actuator circuits should be validated before continuous machine trials.

Applied to this study, Agile hardware development provides the development rationale for co-designing physical mechanics, sensor networks, embedded firmware, GUI controls, and power protection. Rather than delaying firmware and control testing until full assembly, the proposed machine should be developed through repeated subsystem validation. This includes testing the grinding motor, mixing motor, compression actuator, halogen drying control, sensor feedback, GUI process commands, and fault responses in isolated cycles. The framework therefore shifts the development process from a rigid assembly task to a responsive engineering process that can identify faults earlier.

Through this framework, the proposed system is treated not only as a finished machine but as a cyber-physical architecture whose mechanical stability, electrical safety, and logic accuracy should be checked through progressive integration. The Agile hardware development framework supports the system's aim of minimal human intervention because automation is reliable only when the controller, actuators, sensors, and interface operate consistently together. This framework provides the basis for developing, testing, and evaluating the proposed machine as an integrated cyber-physical system.

Synthesis of the Reviewed Literature and Studies

The reviewed literature validates the mechanical and socio-economic necessity of replacing manual solid fuel production with a semi-automated pipeline. Studies confirm that manual briquette production is heavily bottlenecked by physical fatigue, inconsistent molding pressure, and long processing times. The primary strength of this compiled research is its empirical proof that process automation and mechanical densification eliminate non-value-added activities, standardize structural integrity, and optimize combustion while keeping emissions within safe domestic thresholds. Additionally, it establishes that the viability of such systems relies not only on hardware efficiency but also on user acceptability, which must be systematically measured using frameworks like the ISO 25010 quality model.

Anchored on these established frameworks, this research should achieve the complete development of an embedded system architecture for a self-sustaining, solar-powered, and semi-automated paper-charcoal briquette machine. The

engineered pipeline must integrate grinding, soaking, mixing, compression, and halogen-assisted drying to explicitly reduce total man-hour consumption and throughput time compared to the traditional manual baseline. Furthermore, the system must produce standardized, high-density briquettes with measurable shatter resistance and extended burn duration, while providing a highly responsive Graphical User Interface (GUI) to minimize continuous human oversight.

While existing studies frequently investigate isolated variables, such as testing the compressive strength of natural binders, evaluating standalone mechanical presses, or measuring the emissions of specific waste blends, this study synthesizes these fragmented concepts into a single cyber-physical architecture. Unlike prior research that separates mechanical densification from software evaluation, this project bridges the gap by combining physical time-and-motion efficiency with real-time sensor feedback and an interactive GUI. By operating off-grid through an integrated solar panel and conforming to strict material constraints, the proposed system differentiates itself as a fully integrated, community-adoptable technological solution rather than an isolated material experiment.

Chapter 3

METHODOLOGY

Research Design

This study utilizes a quantitative applied developmental research design, specifically anchoring on Type 1 Developmental Research. Following the standard academic taxonomy for developmental studies, Type 1 research is defined as the systematic study of the design, development, and rigorous evaluation of a specific, tangible product or tool. This specific classification is the most appropriate for this study because the primary objective is not to generate generalized theories, but to engineer a functional cyber-physical system (the solar-powered, semi-automated paper-charcoal briquette machine) to solve the practical, real-world inefficiencies of manual briquette production.

The methodology is inherently an applied research approach because it integrates mechanical hardware, low-power subsystems, and a React-based Graphical User Interface (GUI) into a localized solution intended to address a direct community problem. The development process requires strict standardization of material variables (the exact mass ratio of charcoal fines to paper waste and water volume) to ensure a controlled constant during physical extrusion. The machine's architecture relies on the continuous integration of closed-loop feedback sensors such as the YF-S201 for water flow, the HX711 load cell for compression force, and the

DS18B20 for drying-temperature monitoring and DHT22 for enclosure humidity monitoring. These sensors are engineered to replace manual labor and oversight.

Furthermore, a quantitative method is utilized to validate the machine's utility and directly answer the Statement of the Problem. Rather than relying on qualitative observations, the prototype's performance is evaluated using strict numerical and statistical metrics. The system is mathematically tracked to measure the reduction in man-hour consumption, overall throughput time, and standard production rate (units/hr) against an established manual baseline. Finally, the physical output of the machine is quantitatively tested for structural integrity, heat retention, and burning duration, using store-bought wood charcoal as the comparative control. Therefore, this specific research design provides the necessary structured framework to not only construct the embedded architecture but to empirically prove its efficiency.

Flowchart of Research Design

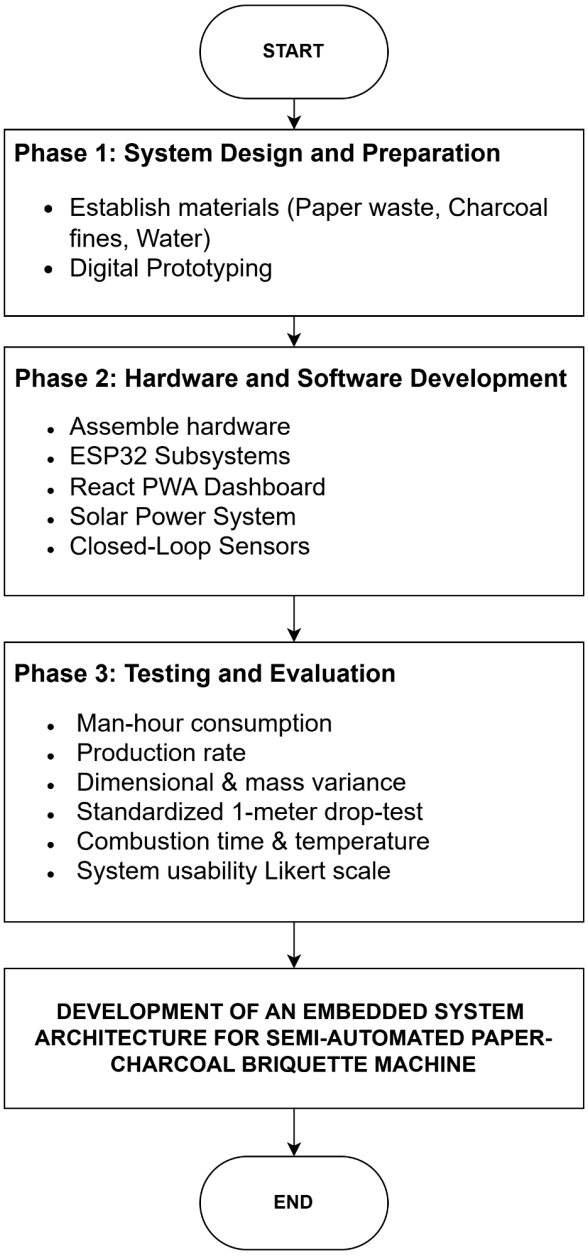


Figure 2. Flowchart of Research Design

The flowchart illustrates the systematic, three-phase developmental pipeline executed in this study. The process initiates with System Design and Preparation,

which focuses on establishing the controlled baseline parameters for the raw materials, specifically the paper waste, charcoal fines, and water. Following the material standardization, the study advances to Hardware and Software Development. This central engineering phase encompasses the physical assembly of the machine's hardware alongside the integration of its technological core, which includes the ESP32 subsystems, the React PWA touch screen for the graphical interface, the off-grid solar power system, and the closed-loop sensory network. Once the cyber-physical prototype is fully constructed, it undergoes Testing and Evaluation. This final stage assesses the system's operational efficiency and physical output through targeted metrics, including man-hour consumption, production rate, dimensional and mass variance, a standardized 1-meter drop-test, combustion time and temperature, and a system usability Likert scale. The successful and sequential execution of these three phases culminates in the finalized output: the complete development of an embedded system architecture for the semi-automated paper-charcoal briquette machine.

Description of Research Instrument Used

Statement of the Problem 1:

To address the first Statement of the Problem regarding the prevailing challenges and inefficiencies in utilizing paper waste and charcoal fines, the study relies on descriptive engineering synthesis and documented literature rather than physical measuring hardware. The primary instrument for this qualitative objective is the analysis of existing waste management and energy demand studies, which

establishes the socio-economic and technical baseline justifying the need for an engineered alternative solid fuel source.

Statement of the Problem 2:

For the second Statement of the Problem, which investigates the physical demands and time consumption of the traditional manual briquette-making process, the primary research instruments are a digital stopwatch and structured observation logs. These tools are utilized to strictly record the continuous manual oversight, human labor hours, and operational bottlenecks of the traditional hand-pressed method, thereby establishing the quantitative baseline required for direct comparison against the semi-automated architecture.

Statement of the Problem 3:

To determine the design and operational limitations of the proposed machine for the third Statement of the Problem, the research utilizes engineering documentation and prototype architectural logs. These descriptive instruments systematically record the mechanical boundaries, safety interlock constraints, and manual material feeding requirements established during the hardware integration phase, ensuring that the system's functional scope and its inability to extract water-insoluble toxins are explicitly mapped and defined.

Statement of the Problem 4:

Addressing the fourth Statement of the Problem regarding the machine's effectiveness in production efficiency and product quality requires a matrix of quantitative hardware instruments. A digital stopwatch is used during live operational runs to track total burning duration, throughput time and man-hour consumption, while

a batch counter logs the standard production rate. Furthermore, an internal 100kg S-Type Load Cell with an HX711 amplifier is utilized to verify consistent mechanical densification, a digital weighing scale and Vernier caliper measure the physical variance of the output, and an infrared thermometer scans radiant temperatures to plot the heat retention of the final briquettes against store-bought charcoal.

Statement of the Problem 5:

Finally, to evaluate the acceptability of the automated system and its Graphical User Interface for the fifth Statement of the Problem, the study utilizes an ISO 25010 Quality Model Questionnaire. This standardized Likert scale evaluation tool is distributed to target users following guided operational runs to quantitatively measure the system's functional suitability, performance efficiency, interface usability, and reliability, specifically focusing on the clarity of the touchscreen prompts, safety faults, and audio-visual cues.

Mix Design

Material Component	Initial Wet Mix Proportion	Final Dried Briquette Proportion
Paper Waste	16.13% - 20%	50%
Residual Charcoal Fines	16.13% - 20%	50%
Water	67.7% - 60%	0%
Total	100%	100%

Table 1. Mix Design Proportions for Paper-Charcoal Briquettes

According to Sjöstrand, B., & Brolinson, A. (2022) in “Addition of polyvinylamine in chemi-thermomechanical pulp and kraft pulp and the effects on dewatering,

strength, and air permeance". Kraft pulp, the same material used to make the common paper bags, has the Water Retention Value (WRV) of 1.0-1.2, which is the ability of paper to hold water even through compression. According to Gupta, S. S., Prasad, R., Chakraborty, D., & Ebbing, M. (2026), the relevant Water Holding Capacity, the ability to hold water, of charcoal fines to the study is 2.0-3.0. The goal ratio of the mixture of research between paper and charcoal is 1:1, therefore if the paper of the same weight of the charcoal fines undergoes the process of soaking, then the ratio of the mixture after molding and compression would be 1.0:1.0:3.0 to 1.0:1.0:4.2 where charcoal fines:paper:water respectively. This will conclude the composition of the wet mixture to be:

$$\%paper = \frac{1}{5} \times 100 = 20\%; \%paper = \frac{1}{6.2} \times 100 = 16.13\%$$

$$\%charcoal = \frac{1}{5} \times 100 = 20\%; \%charcoal = \frac{1}{6.2} \times 100 = 16.13\%$$

$$\%water = \frac{3}{5} \times 100 = 60\%; \%water = \frac{4.2}{6.2} \times 100 = 67.7\%$$

The water content in the dried briquette should be negligible, therefore it will be 0%.

Data Gathering Procedure

The data gathering procedure will formally commence with the identification and selection of the target beneficiaries, specifically fifteen representative small-scale end-users who have no prior experience operating the machine. Before any testing begins, the researchers will secure formal permission and obtain informed consent from these participants, ensuring they are fully aware of the study's purpose and their role in the evaluation process. This step is critical to ensure ethical compliance and to establish a cooperative environment for the subsequent instrumentation phases.

To determine the existing production constraints, the researchers will first gather baseline data directly from the participants as they perform the traditional manual briquette-making process. During this stage, the researchers will utilize digital timers and observation logs to conduct a time-and-motion study, strictly tracking the participants' total man-hour consumption and throughput time. This procedure yields the quantitative baseline data required to measure their current physical demands and operational efficiency before any technological intervention is introduced.

Following the manual baseline observation, the researchers will orient the participants on the operation of the semi-automated paper-charcoal briquette machine. The participants will then be asked to perform a guided production task, operating the system exclusively through the touch screen Graphical User Interface (GUI). This allows the researchers to gather live time-and-motion data on the automated throughput and directly observe how effectively the users interact with the machine’s process confirmations, safety fault alerts, and audio-visual cues.

Concurrently, as the participants operate the machine, the researchers will strictly record the necessary experimental data regarding the prototype's physical output. A sample of thirty machine-produced briquettes will be systematically collected from the users' batches for physical and volumetric variance tracking using a digital weighing scale and Vernier caliper. Furthermore, the researchers will execute the standardized one-meter drop-tests and thermal combustion tests using an infrared thermometer to quantitatively gather the structural and thermal performance data of the briquettes.

Upon the completion of the physical production cycle, the final phase of data gathering involves the direct administration of the survey instrument to the participants. The researchers will distribute the standardized usability questionnaire to the fifteen target users to assess their firsthand experience regarding the system's functional suitability, interface learnability, and overall reliability. The researchers will assist any participants who require clarification regarding the survey items, after which the completed questionnaires will be collected, tallied, and subjected to the designated statistical treatments to finalize the evaluation of the system..

Material Requirements

Central Processing, Network Routing, and Memory Units

Component	Image	Description	Application
ESP32-WROOM-32D Dual-Core Core		A 32-bit symmetric multiprocessing architecture with dual execution cores.	Core 0 manages low-latency hardware ISRs and fail-safe interlocks. Core 1 handles the local networking stack, Wi-Fi AP, and 10 Hz WebSocket telemetry streaming.

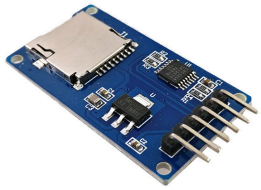
Nextion 4.3” Smart UART Touchscreen LCD Assembly		Intelligent HMI panel with onboard flash memory and an independent graphics processor.	Uses a dedicated serial bus to display real-time data and user verification panels, capturing inputs without burdening the master CPU.
Physical SPI Micro-SD Card Module		Non-volatile edge-storage module interfacing via a high-speed external VSPI bus.	Ensures research data persistence by writing session arrays from the RAM cache to a physical FAT32 partition upon cycle completion.

Table 2. Central Processing, Network Routing, and Memory Units

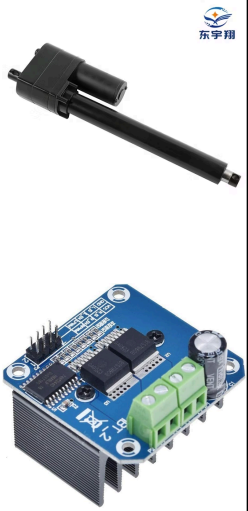
Autonomous Off-Grid Photovoltaic Power Infrastructure

Component	Image	Description	Application

100W Monocrystalline Photovoltaic Array		High-efficiency solar harvesting panel for varying irradiance.	Primary energy source; provides DC charging to maintain system power balance.
10A Intelligent MPPT Solar Charge Controller		Multi-stage battery regulator using MPPT algorithms.	Optimizes charging voltage; protects against over-current and over-charge.
12V 30Ah LiFePO4 Battery Bank		High-density battery with superior thermal stability and long lifecycle.	Energy reservoir; supports high-current compaction cycles without brownouts.
LM2596S DC-to-DC Step-Down Buck Converters		High-efficiency switching step-down power regulators.	Converts 12V bus to stable 5V and 3.3V rails for HMI and logic circuits.

Table 3. Autonomous Off-Grid Photovoltaic Power Infrastructure

Electromechanical Actuation, Fluidics, and Curing Modules

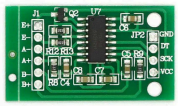
Component	Image	Description	Application
Heavy-Duty Linear Actuator & BTS7960 Driver Matrix		High-torque industrial ram paired with a high-current H-bridge driver.	Primary compaction engine to compress slurries into briquettes and used as control arms for trapdoors, controlled via PWM for precise velocity and torque.
12V 775 Pulverizer and Geared Planetary Mixing Motors		Dual-motor configuration: high-speed DC motor and low-RPM, high-torque planetary geared drive.	High-speed motor pulverizes paper waste; geared motor drives agitation blades for homogeneous blending of wet slurry.

Solenoid-Driven Fluid Distribution Manifold		Electronic fluidics system with a solenoid valve and a motorized brass ball valve.	Solenoid valve injects water; motorized valve flushes excess water extracted from consolidated paper slurry.
MG996R Core High-Torque Servo Motor		High-torque rotary position actuator with a reinforced metal gear train.	Actuates internal sliding barrier tracks to dispense dry charcoal dust increments under automated PWM control.
12V ERM Slurry-Settling Vibration Array		Electric vibration motor configured with an off-center balanced Eccentric Rotating Mass (ERM) assembly.	Executes a brief high-frequency vibration routine in the mold cavity to normalize slurry density before compaction.

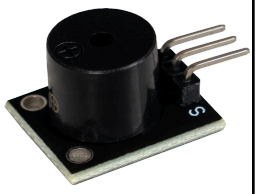
50W Short-Wave Infrared Halogen Array & Forced Airflow System	 	Environmental curing configuration combining a four-element infrared halogen array with a brushless DC fan.	Halogen lamps deliver thermal energy for curing; fan vents moisture-saturated air out of the enclosure during drying.
Multi-Channel Optocoupled Logic Relay Channels		Array of optically isolated solid-state electromagnetic switching relays.	Solid-state power isolation gateway, physically isolating high-power motor and heating coil loops from the CPU logic traces.

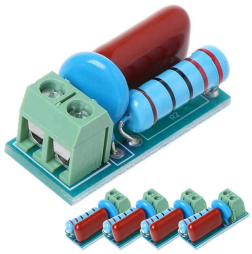
Table 4. Electromechanical Actuation, Fluidics, and Curing Modules

Closed-Loop Sensory Feedback and Fail-Safe Interlocks

Component	Image	Description	Application
S-Type Structural Load Cell & HX711 Instrumentation Amplifier	 	Alloy steel strain-gauge sensor with high-resolution 24-bit ADC.	Monitors and transmits digital data to the CPU to confirm when target compaction force (≥2000N) is reached.
YF-S201 Hall-Effect Liquid Flow Sensor		Inline counting device with magnetic impeller and Hall-effect IC.	Tracks liquid infusion via hardware interrupts to maintain controlled water dosing during soaking and mixing.

XK-Y25 Non-Contact Capacitive Liquid Level Sensor		Non-invasive proximity sensor for reading fluid dynamics through pipe walls.	Detects liquid presence via capacitive variance to verify drainage before mixing phases.
DS18B20 1-Wire Waterproof Thermal Probes		Waterproof digital sensors using Dallas 1-Wire serial protocol.	Three-probe redundancy array for tracing internal environment heating curves and safety targets.
DHT22 Temperature & Humidity Sensor		Calibrated digital sensor for ambient temperature and relative humidity.	Triggers automated cycle termination when exhaust humidity reaches ≤15% Relative Humidity (RH).

HC-SR04 Ultrasonic Distance Board Matrix		Ranging transducers using acoustic pulse time-of-flight wave properties.	Eight-module array provides a spatial matrix to prevent cycles if storage levels are full.
Roller Lever Micro Interlock Switches		Active-low physical contact ground loop micro-switches.	Safety interlock that cuts actuator output within 19.2ms if mechanical enclosures are opened.
Active Piezo Signalling Buzzers		High-decibel electronic modules generating fixed acoustic frequencies.	Decentralized three-module array for signaling process boundaries and manual operator prompts.

Emergency Shutdown Switch		Heavy-duty, lockable Red Mushroom E-Stop Button	Manual safety switch to instantly cut off the main 12V battery power line
Inductive Snubbers		Transient voltage suppression circuits (typically flyback diodes or RC networks).	Placed across inductive loads to safely dissipate high-voltage spikes, preventing electromagnetic interference
Electrical Enclosure		An IP65 Waterproof plastic control panel box	Protects the ESP32, motor drivers, buck converter, and relays from water splashes.
Logic Filter Capacitors		Ceramic and electrolytic decoupling (bypass) capacitors.	Placed across the VCC and GND pins of the ESP32 and sensitive ADC modules to filter out

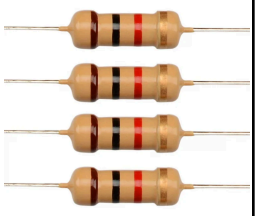
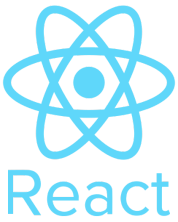
			high-frequency power supply noise and stabilize sudden voltage drops
Resistors		Passive electrical components utilized for signal conditioning and current limiting.	Configured as pull-up or pull-down networks for the micro interlock switches and 1-Wire communication lines (DS18B20) to ensure stable logic levels

Table 5. Closed-Loop Sensory Feedback and Fail-Safe Interlocks

Software Requirements

Name	Image	Description	Application
React.js + Vite (PWA)		Component-based JS library bundled for performance via Vite.	Primary HMI and Telemetry HUD, renders timers, warnings, and dashboards via offline GZIP files.

FreeRTOS (C++ Firmware)		RTOS supporting multi-threading, prioritization, and strict timing constraints.	ESP32 kernel; uses dual-core threading to separate safety ISRs (Core 0) from UI/network tasks (Core 1).
ESPAsyncWebServer & WebSockets		Asynchronous HTTP server and full-duplex communication protocol.	Manages local Wi-Fi AP and serves PWA; maintains 10Hz WebSocket telemetry for real-time UI updates.

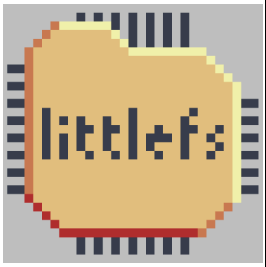
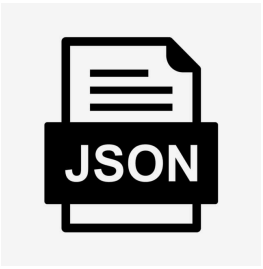
LittleFS (Internal Storage)		Fail-safe filesystem for MCUs, featuring wear leveling and power-loss resilience.	Stores PWA assets (HTML/CSS/JS) on ESP32 internal flash for instant offline dashboard loading.
FAT32 Micro-SD & CSV Logger		Standard file system for removable media; stores structured data in CSV format.	Local edge database; logs runtime, water mass, and compaction force per batch for CSV download.
ArduinoJson		JSON serialization/deserialization library optimized for embedded C++.	Packages hardware telemetry (current, loads) into JSON strings for network broadcasting.

Table 6. Software Requirements

Statistical Treatment

Statement of the Problem 2:

To address the second Statement of the Problem regarding the physical demands and time consumption of the manual baseline, the study utilizes the arithmetic mean. This statistical tool calculates the average throughput time and human labor consumed across multiple manual production trials by dividing the sum of all recorded baseline data points by the total number of manual trials. Establishing this mean provides a reliable, mathematical baseline necessary for direct comparative analysis against the automated machine.

$$\bar{x} = \frac{\Sigma x_i}{n}$$

(Where x is the mean, xi is the sum of all recorded data points, and n is the total number of trials).

Statement of the Problem 4:

To determine the machine's effectiveness in improving production efficiency for the fourth Statement of the Problem, specific operational formulas are applied to the gathered time-and-motion data. The percentage reduction formula is utilized to explicitly quantify the exact proportion of human labor and overall processing time eliminated by transitioning to the semi-automated pipeline. Concurrently, the standard production rate formula evaluates the machine's throughput by calculating its standardized hourly yield, which is derived by dividing the total number of successfully produced briquettes by the total operational time in hours.

$$\bar{x} = \frac{\Sigma x_i}{n}$$

(Where x is the mean, xi is the sum of all recorded data points, and n is the total number of trials).

$$\%Reduction = \frac{T_{manual} - T_{semi-automated}}{T_{manual}} \times 100$$

(Where Tmanual is the recorded time/labor value of manual production, and Tsemi-automated is the recorded time/labor value of the machine).

$$R = \frac{U}{T}$$

(Where R is the production rate in units/hr, U is the total number of briquettes produced, and T is the total operational time in hours).

Further addressing the fourth Statement of the Problem concerning product quality and fuel performance, the study employs both the arithmetic mean and standard deviation. The arithmetic mean calculates the average burning duration and heat retention temperatures across all comparative test trials against the store-bought charcoal control. Meanwhile, the standard deviation is utilized to measure the physical consistency of the machine's output. By evaluating the mass and dimensional variance of the sampled briquettes, a lower standard deviation statistically proves that the machine successfully mitigates the physical variability inherent in traditional hand-molding.

$$\bar{x} = \frac{\Sigma x_i}{n}$$

(Where x is the mean, xi is the sum of all recorded data points, and n is the total number of trials).

$$\sigma = \sqrt{\frac{\Sigma(x_i - \overline{x})^2}{n}}$$

(Where is the standard deviation, xi is each individual briquette's value, x is the mean, and n is the population size).

Furthermore, to evaluate the structural integrity of the output for the fourth Statement of the Problem, the shatter resistance index or percentage weight loss formula is applied. Following a standardized drop-test, this formula calculates the percentage of mass lost by subtracting the weight of the largest remaining solid piece from the initial weight of the briquette, dividing that difference by the initial weight, and multiplying the quotient by one hundred. This specific mathematical treatment explicitly quantifies the physical durability of the final product, providing hard data to prove whether the machine-pressed briquettes can successfully resist shattering upon impact compared to standard charcoal.

$$\%Weight\ loss = \frac{W_i - W_o}{W_i} \times 100$$

(Where Wi is the initial weight of the briquette before the drop, and Wo is the weight of the largest remaining solid piece after the drop).

Finally, to evaluate the acceptability of the semi-automated system and its Graphical User Interface for the fifth Statement of the Problem, the study utilizes the weighted mean. This statistical treatment processes the distribution of responses gathered from the 5-point Likert scale ISO 25010 questionnaire. By dividing the sum of the products of the frequencies and their corresponding weights by the total number of respondents, the resulting weighted mean is matched against a

standardized verbal interpretation scale to quantitatively conclude the system's functional suitability, performance efficiency, usability, and reliability.

$$\bar{x} = \frac{\Sigma x_i}{n}$$

(Where x is the mean, xi is the sum of all recorded data points, and n is the total number of trials).

Design Project Flow

The Design Project Flow outlines the engineering execution required to build the semi-automated paper-charcoal briquette machine. This section details the integration of the mechanical hardware, low-power microcontrollers, and software logic that drives the continuous production pipeline. To ensure a reproducible development cycle, the system design is structured into three core components: the Block Diagram, the System Architecture, and the Prototype Design.

Block Diagram

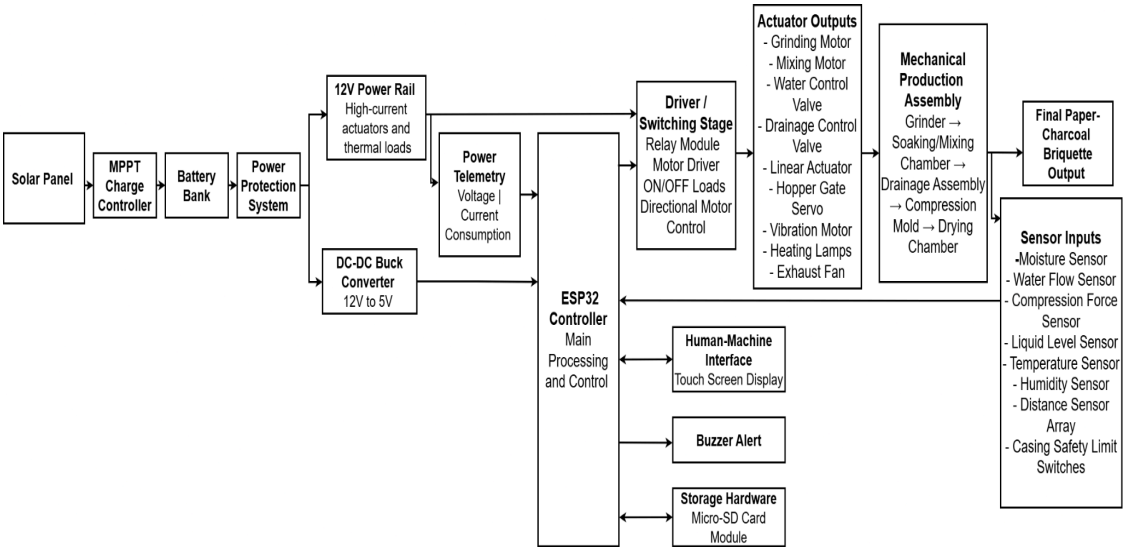


Figure 3. Block Diagram of the Hardware Flow

The block diagram shows the hardware flow of the semi-automated paper-charcoal briquette production system with graphical user interface. The power section starts from the 100W monocrystalline solar panel, which is regulated by the MPPT charge controller and stored in the 12V 30Ah LiFePO4 battery bank. The battery output passes through the emergency stop and fuse system before being distributed to the 12V power rail for high-current loads and to the DC-DC buck converter for the ESP32 controller and low-voltage components.

The 12V power rail supplies the actuator section, including the grinding motor, mixing motor, water control valve, drainage control valve, linear actuator, hopper gate servo, vibration motor, heating lamps, and exhaust fan. The relay module controls the ON/OFF loads, while the IBT-2 BTS7960 motor driver controls the linear actuator. These actuators perform the main production process: grinding, soaking/mixing, drainage, compression molding, and drying.

The ESP32 controller receives feedback from the sensor inputs, including the moisture sensor, YF-S201 water flow sensor, S-Type load cell with HX711 amplifier, liquid level sensor, DS18B20 temperature sensor, DHT22 humidity sensor, HC-SR04 ultrasonic distance sensor array, and casing safety limit switches. These sensors monitor water flow, moisture, compression force, liquid level, drying temperature, humidity, distance, and safety conditions during operation. Power telemetry monitors voltage, current, and consumption through the INA219 current/power monitoring module.

The ESP32 also connects to the Nextion HMI touchscreen, buzzer alert, and micro-SD card module. The touchscreen provides user control and system monitoring, the ESP32 activates the buzzer to provide audio alerts for process notifications and system faults, and the micro-SD card stores batch logs and configuration files. Overall, the block diagram shows how the power supply, protection system, controller, sensors, actuators, interface, and storage hardware are connected to produce the final paper-charcoal briquette output.

Process Flowchart

To ensure deterministic control and maintain the required human-in-the-loop safety framework, the system's operational logic is divided into five distinct, modular phases governed by the ESP32 microcontroller's finite state machine. The overarching process flow dictates the sequence of electromechanical actuation, sensor polling, and human-machine interface (HMI) authorizations.

Module 1: System Initialization and Diagnostics

The system begins in the Initialization state upon receiving power from the solar-battery infrastructure.

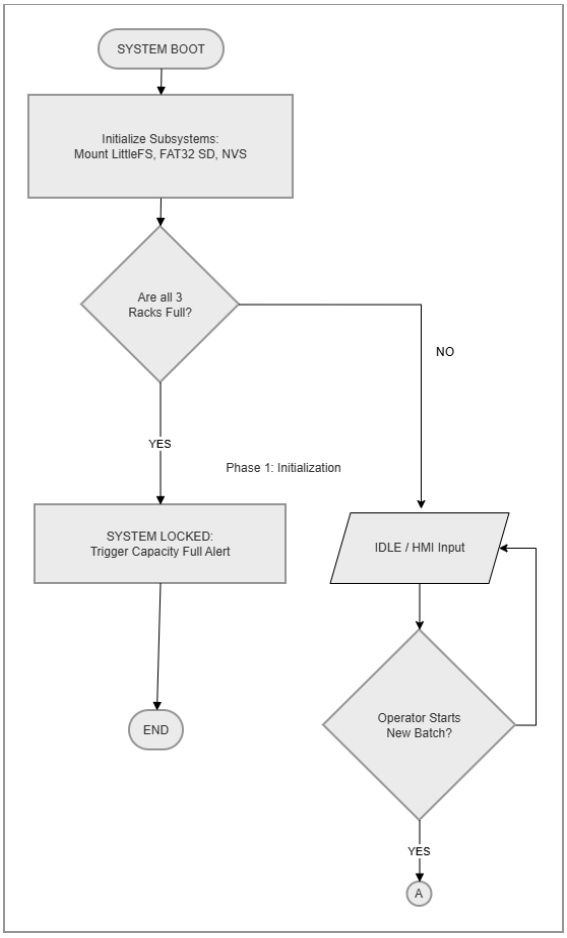


Figure 4. Process Flow: Initialization and Diagnostics

Figure 4 illustrates the initial boot sequence and diagnostic protocols of the ESP32 microcontroller's finite state machine. Upon receiving power from the solar-battery infrastructure, the system immediately initializes its low-power subsystems by mounting the internal LittleFS flash partitions and external FAT32 SD card modules, ensuring the NVS (Non-Volatile Storage) registry is booted for state

retention. Concurrently, the dual-core processor establishes the local network infrastructure by broadcasting a Wi-Fi Access Point and hosting the responsive React PWA dashboard assets, successfully preparing the centralized human-machine interface (HMI) for operator interaction.

Module 2: Grinding and Soaking Phase

This phase prepares the raw composite materials for integration.

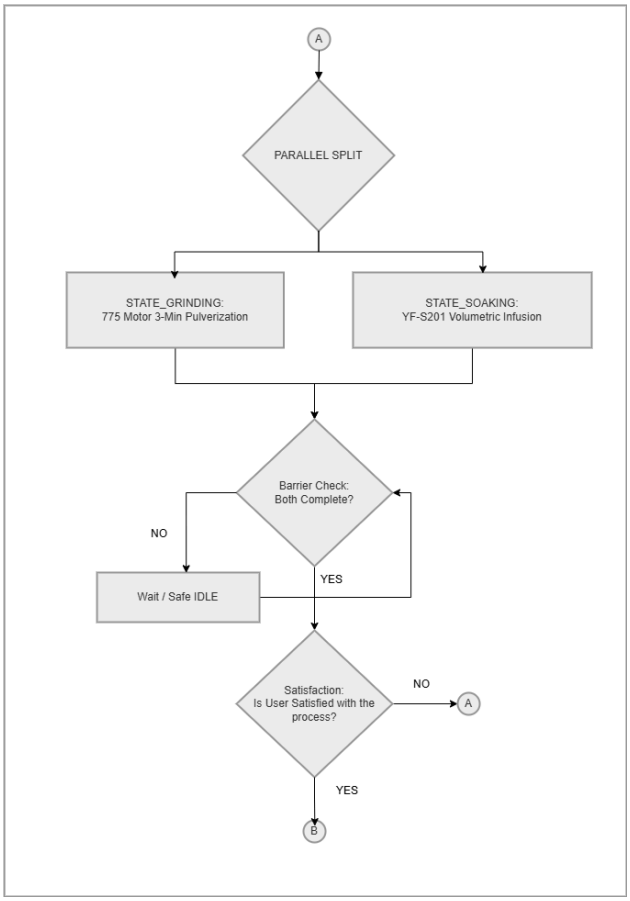


Figure 5. Process Flow: Grinding and Soaking State

Figure 5 delineates the logic flow for the simultaneous preparation of the raw biomass materials. The algorithmic sequence dictates the actuation of the 12V high-speed DC motor relay to pulverize residual charcoal, while actively monitoring the

motor's operating voltage and amperage via an asynchronous I2C power sensor. In parallel, the soaking process is executed by shifting a normally closed solenoid water valve to an open state. The microcontroller utilizes a microsecond hardware interrupt service routine tied to the YF-S201 Hall-Effect sensor to measure the volumetric inflow, looping this process until the precise 1:1 water-to-paper target ratio is achieved.

Module 3: Combining Phase

This module executes the homogenization of the paper-water slurry and the pulverized charcoal.

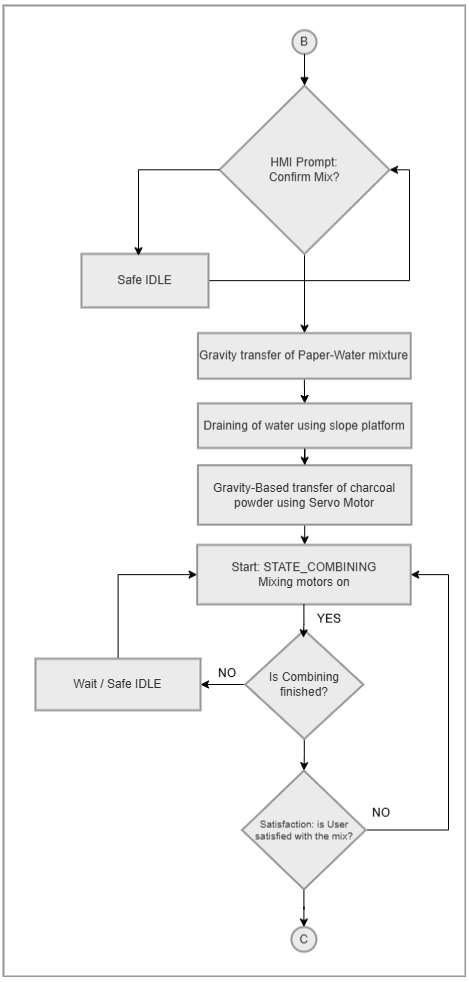


Figure 6. Process Flow: Mixing State

Figure 6 details the algorithmic progression of the Combining Phase, where the hydrated paper slurry and pulverized charcoal fines are mechanically integrated. Following strict human-in-the-loop protocols, this state only executes after receiving explicit operator confirmation via the touchscreen HMI. Once authorized, the ESP32 drives the 12V planetary geared blending motor relay. The system loops through a predefined mixing timer, ensuring continuous and consistent homogenization of the composite matrix, before triggering active buzzer pulses to prompt the operator for the next stage transition.

Module 4: Compression and Molding Phase

This is the core densification stage requiring strict physical sensor and actuator oversight.

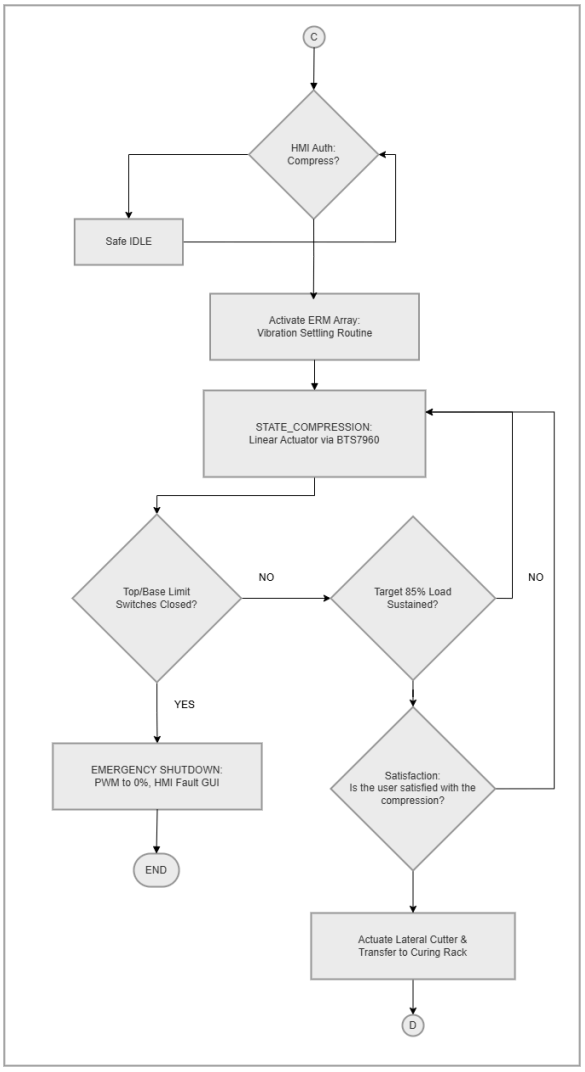


Figure 7. Flowchart: Compression and Molding Stage

Figure 7 maps the critical control logic and fail-safe interlocks during the core densification stage. Because this phase utilizes a heavy-duty linear actuator driven by a BTS7960 motor driver, the ESP32 continuously polls the hardware safety limit switches. If a casing breach is detected, the flowchart branches into an emergency shutdown protocol, immediately dropping the actuator duty cycle to 0%. Under normal operation, the system drives the actuator downward while actively polling the HX711

24-bit ADC and S-Type load cell, halting displacement only when the target structural compaction load of 2000N is successfully sustained.

Module 5: Thermal Curing (Drying) and Data Logging

The final phase controls moisture reduction and edge-heavy data logging.

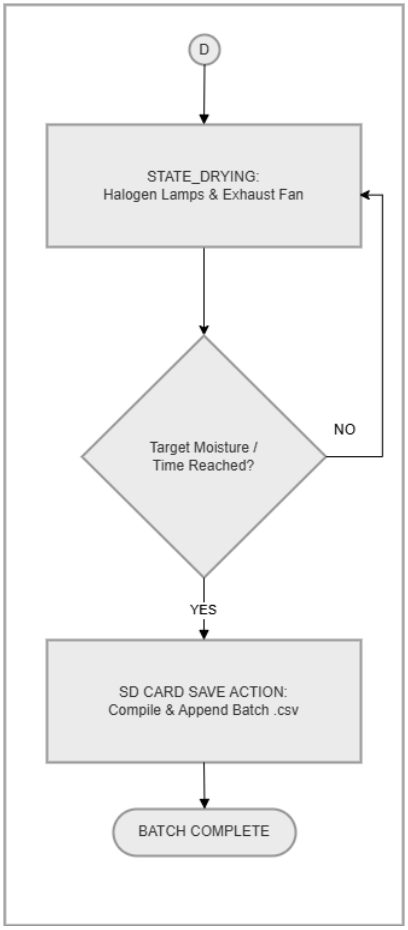


Figure 8. Flowchart: Drying Stage

Figure 8 outlines the final thermal curing logic and the edge-heavy data logging operations. Upon operator authorization, the system energizes the infrared halogen thermal arrays and the exhaust ventilation blower relay. As the system loops, it continuously polls the DS18B20 thermal probes and the DHT22 relative humidity

sensor. During this cycle, telemetry data strings are compiled and appended directly onto the standalone SD card as a CSV run log. The active drying state is programmed to terminate autonomously only once the exhaust boundary relative humidity drops to or below 15%, completing the production cycle and clearing the active session's transient memory cache.

System Architecture

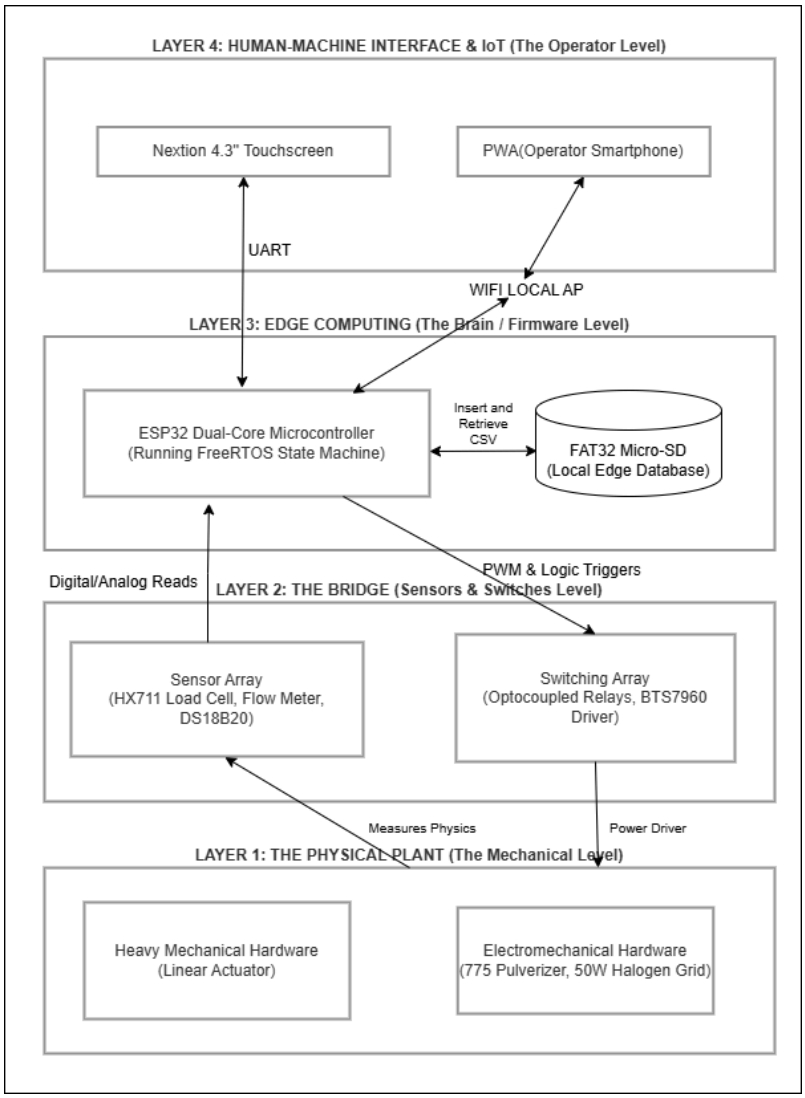


Figure 9: System Architecture of the Semi-Automated Paper-Charcoal Briquette Machine

The architecture utilizes a closed-loop control system driven by an ESP32 dual-core microcontroller running a FreeRTOS state machine. The system integrates real-time sensor feedback, parallel processing, and an local edge framework to ensure consistent production quality, operator safety, and autonomous data logging.The system architecture is divided into four distinct hierarchical layers:

Layer 1: The Physical Plant (Mechanical Level): Comprises the heavy electro-mechanical hardware, including a 12V 775 grinder motor, a high-torque planetary mixer, a linear actuator for compression, an ERM vibration motor for slurry settling, and halogen thermal arrays for drying. It is powered by an independent 12V solar infrastructure (100W Panel, MPPT Controller, and 30Ah LiFePO4 Battery).

Layer 2: The Bridge (Sensor & Switching Level): Acts as the intermediary interface. An array of sensors (YF-S201 flow meter, HX711 load cell, capacitive liquid level, DS18B20/DHT22 climate sensors, and safety limit switches) captures physical metrics. Conversely, an optocoupled relay module and BTS7960 motor drivers isolate and route 12V power to the actuators based on 3.3V logic signals.

Layer 3: Edge Computing (Firmware Level): The ESP32 microcontroller acts as the central brain. It handles parallel task execution, sensor data processing, logic gating, and localized data storage via a FAT32 Micro-SD card module for batch metric logging.

Layer 4: Human-Machine Interface (Operator Level): Operator interaction is facilitated through two primary interfaces: a localized Nextion 4.3" Touchscreen (via UART) and a React Progressive Web App (PWA) hosted on the ESP32's local Wi-Fi Access Point, providing real-time remote monitoring and manual override capabilities.

Prototype Design

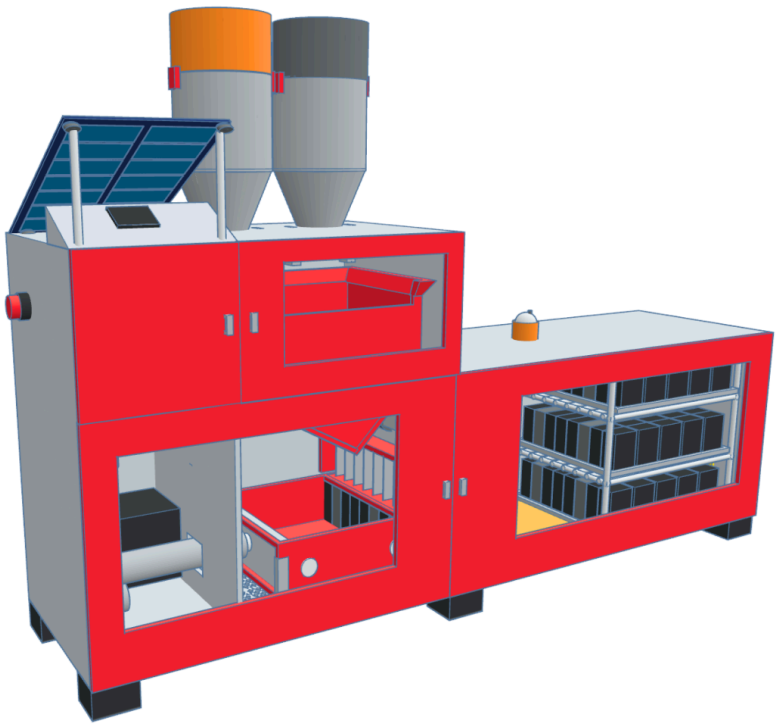


Figure 10. Isometric View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine

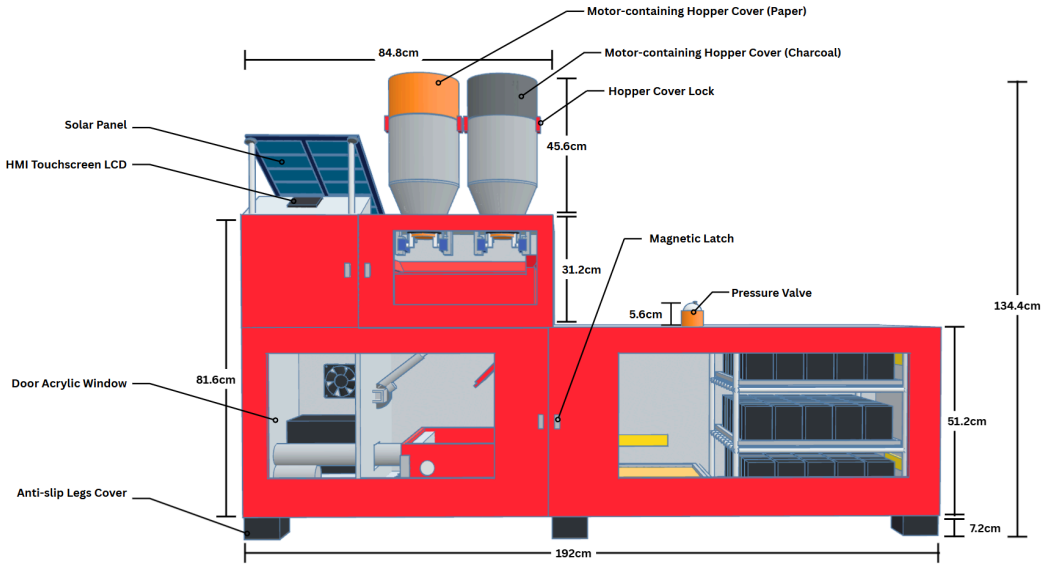


Figure 11. Front View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine

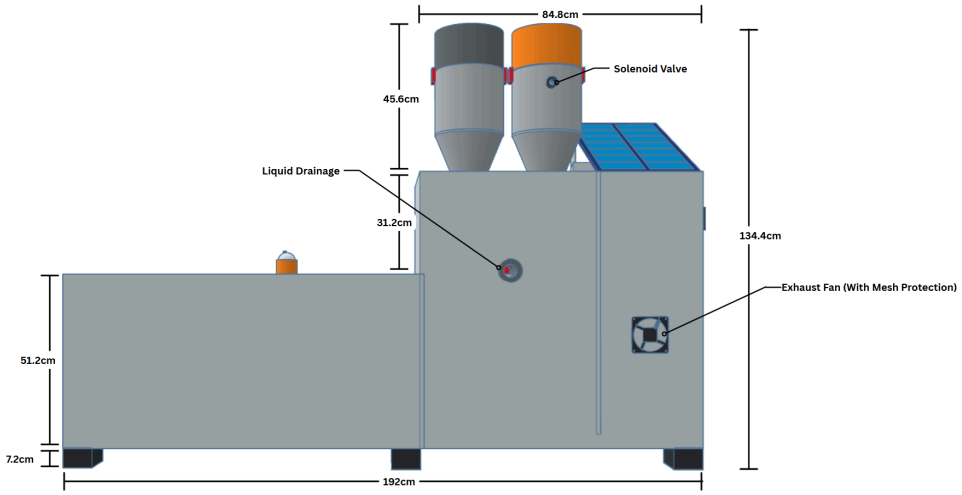


Figure 12. Rear View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine

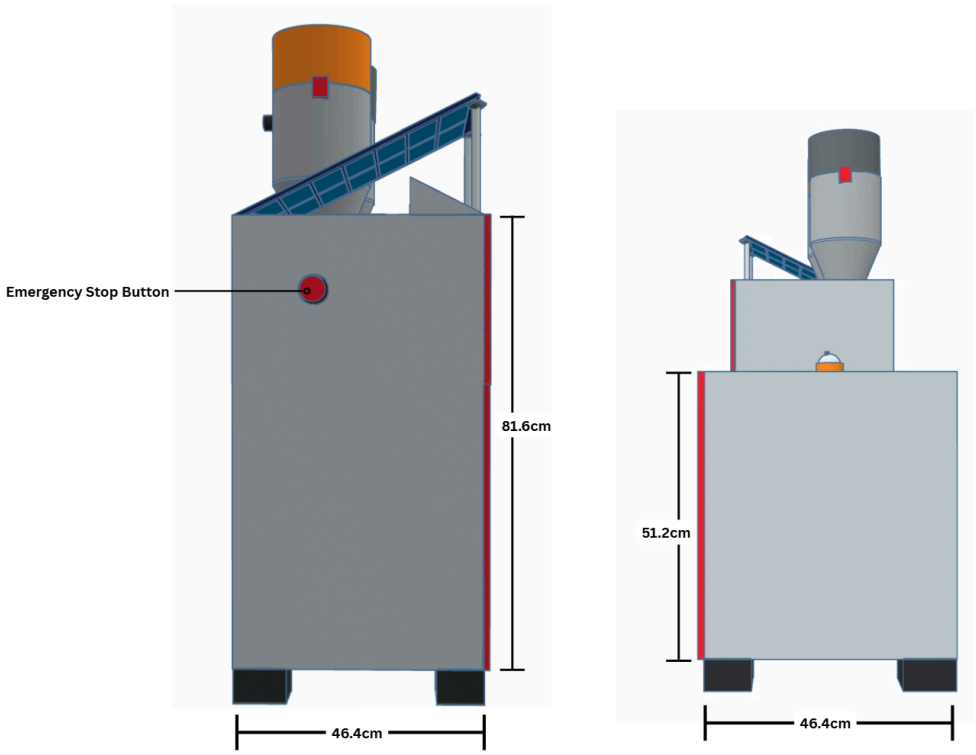


Figure 13. Left and Right Side Views of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine

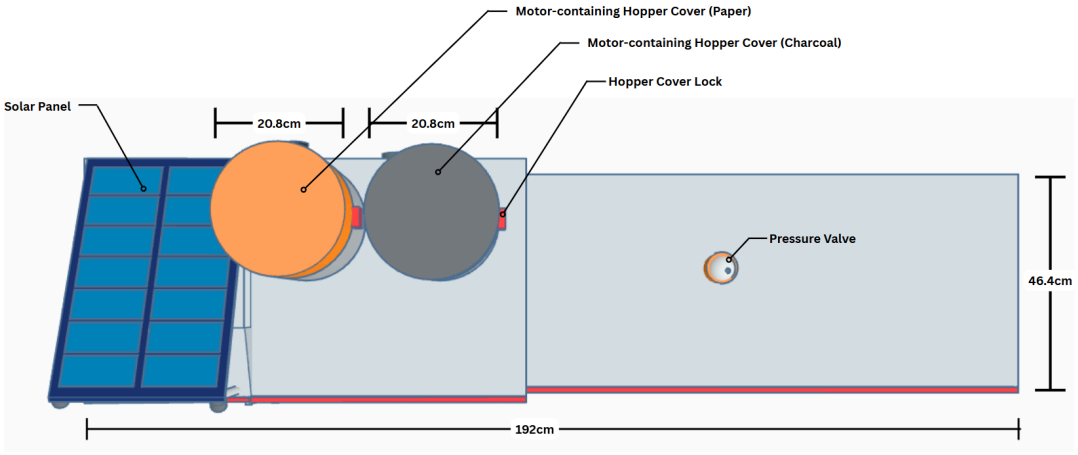


Figure 14. Top View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine

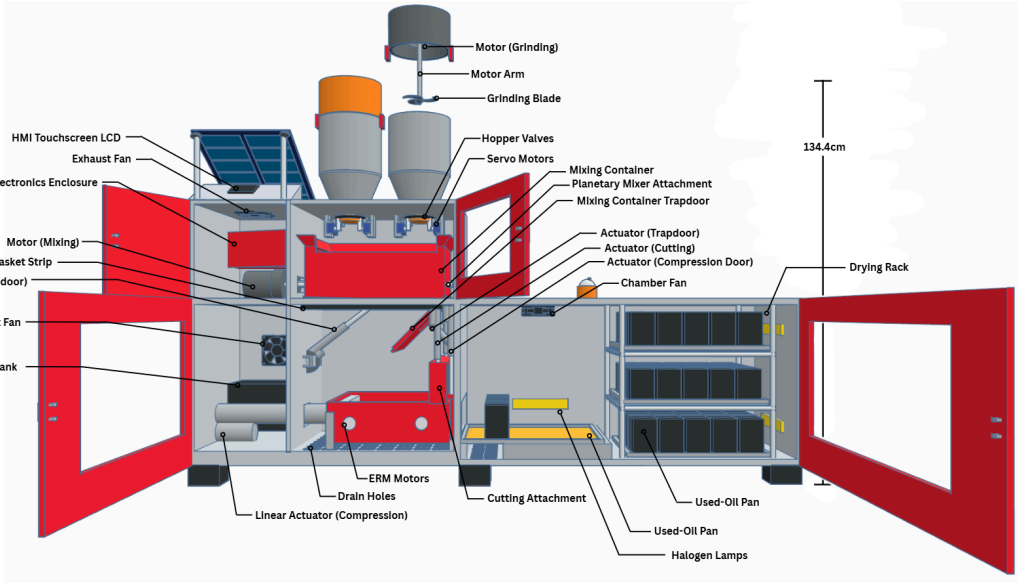


Figure 15. Internal Mechanisms and Layout View of the Prototype Design for the Semi-Automated Paper-Charcoal Briquette Machine

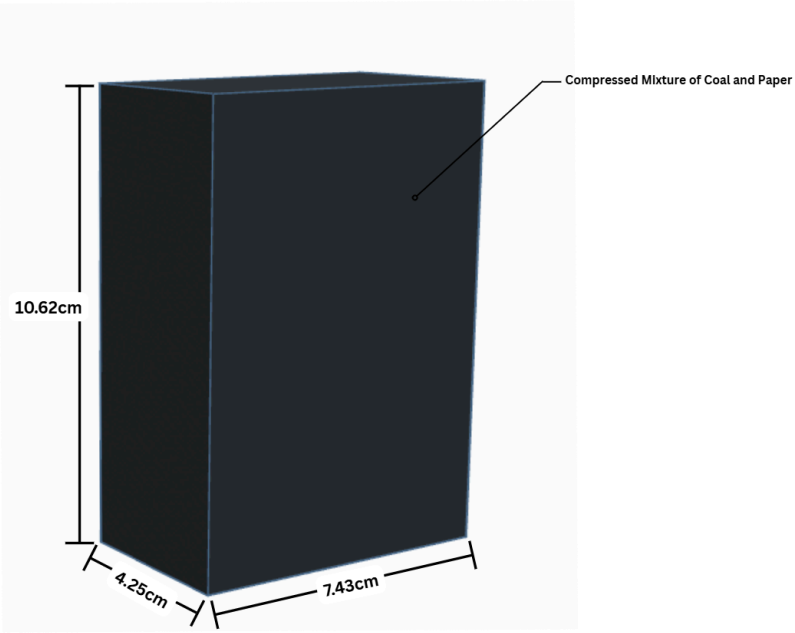


Figure 16. Expected Output Briquette Size

Specimen Details

To systematically conduct the research instruments, the evaluations should be conducted with maintaining the 3 specimens consistently, mainly the Primary test specimens, Raw material specimens, and Control specimens.

The tests to be done to answer the statement of the problems concerns the 30 end products produced by the semi-automated paper-charcoal briquette production. These specimens will undergo evaluation on dimensions, structure, and chemical ability. The stated evaluation includes: dimensional variance tracking, mass measurement, standardized drop test, burning time measurement, and burning temperature measurement. These 30 specimens stated above are the Primary test specimens.

Raw material specimens on the other hand, directly affects the primary test specimens as they are the inputs for the semi-automated paper-charcoal briquette production.. These are strictly limited to: charcoal fines, waste paper, and water.

Control specimens should be the baseline for the performance of the end products of the semi-automated paper-charcoal briquette production and should be surpassed by the new technology. These are the common store-bought charcoal and the manually produced paper-charcoal briquette.

Multiple Constraints

The execution of this developmental research and the operation of the semi-automated machine are strictly bounded by the following physical, operational, and material constraints:

1. Environmental and Power Constraints

Constraint Name	Description
Off-Grid Infrastructure Constraint	To ensure deployment viability in resource-constrained areas, the entire hardware architecture is constrained to operate entirely off-grid, drawing its operational energy exclusively from the integrated solar panel and its associated battery bank.
Weather Dependency Constraint	The system's power generation is strictly dependent on local solar irradiance. Prolonged overcast weather will degrade the charging efficiency of the 100W panel, potentially depleting the 12V 30Ah battery and halting heavy operations (e.g., thermal curing) until recharged.

Table 7. Environmental and Power Constraints

2. Material and Output Constraints

Constraint Name	Description
Material Constraint	The semi-automated pipeline is restricted to processing exactly three input materials: paper waste, residual charcoal fines, and water.
Impurities Constraint	The system is incapable of extracting any water-insoluble impurities in the paper waste prior to the soaking phase.
Shape Constraint	The physical geometry of the final output is strictly limited to a

	solid rectangular prism designed specifically for cooking.
Size Constraint	The internal molding and linear actuator assembly is engineered to produce a single, standardized briquette size.

Table 8. Material and Output Constraints

3. Process and Mechanical Constraints

Constraint Name	Description
Batch Operation Constraint	The system is constrained to a discrete batch-processing workflow. A new batch of raw materials cannot be loaded or processed until the current briquette has completed its entire manufacturing cycle and has been physically removed from the mold.
Viscosity and Torque Constraint	The blending capability is limited by the maximum stall torque of the mixing motor. The system relies on the operator to maintain a reasonable paper-to-water ratio; an excessively dry or dense slurry may stall the motor or trigger current overdraws.
Actuator Stroke Constraint	The fixed stroke length of the linear actuator limits the maximum volume of wet slurry that can be compressed in a single cycle without risking mechanical binding.
Compaction Limit	The densification of the briquettes is strictly hard-capped by the

Constraint	maximum mechanical thrust of the installed linear actuator and the structural integrity of its localized mounting brackets.
Thermal Curing Constraint	The active drying phase is limited by the maximum thermal output of the 50W halogen lamps; therefore, total moisture reduction time remains partially dependent on external ambient humidity and baseline environmental temperatures.

Table 9. Process and Mechanical Constraints

4. Hardware and Sensor Constraints

Constraint Name	Description
Vibration Interference Constraint	The precision of the HX711 load cell is constrained by physical noise; heavy vibrations generated by the pulverizer and mixing motors may induce minor reading fluctuations during simultaneous motor operation.
Volumetric Resolution Constraint	The automated water intake is constrained by the physical resolution of the flow meter, limiting the system from dispensing precise microliter-level measurements.
Thermal Sensing Constraint	The DS18B20 temperature probes measure localized ambient temperature within the drying enclosure, rather than the internal core temperature of the molded briquettes themselves.

Table 10. Hardware and Sensor Constraints

5. Embedded System and Interface Constraints

Constraint Name	Description
Microcontroller Processing Constraint	The system is constrained to the computational limits of the ESP32-WROOM-32D, utilizing a dual-core architecture to strictly separate asynchronous network tasks from deterministic physical control logic to prevent system latency.
Storage Media Constraint	Localized data logging is strictly limited to an external FAT32 formatted micro-SD card utilizing an append-only Comma-Separated Values (CSV) structure, as the microcontroller lacks the RAM required to run standard relational database engines (SQL).
HMI Display Constraint	The physical Human-Machine Interface (HMI) is constrained to a 4.3-inch touchscreen, limiting the on-machine rendering of complex, high-density analytical graphs and requiring the PWA for deeper data extraction.

Table 11. Embedded System and Interface Constraints

6. Software and Network Constraints

Constraint Name	Description
Network	The system operates strictly as a localized Edge IoT node; it

Connectivity Constraint	does not possess cellular or wide-area network (WAN) connectivity, entirely preventing the transmission of telemetry data to remote internet or cloud servers.
Wireless Client Constraint	The localized Software Access Point (SoftAP) broadcasted by the microcontroller is constrained to reliably support a maximum of 4 to 5 simultaneous connected mobile devices for monitoring the Progressive Web App (PWA) dashboard.
Web Application Dependency Constraint	Because the system operates entirely off-grid, the PWA must be fully self-contained within the microcontroller's internal LittleFS partition, explicitly prohibiting the use of external Content Delivery Networks (CDNs) for fonts, icons, or CSS frameworks.

Table 12. Software and Network Constraints

7. Human-Machine Operational Constraints

Constraint Name	Description
Loading Constraint	The system relies on the operator to manually feed the raw biomass materials (shredded waste paper and charcoal fines) into the designated hoppers, as the machine lacks an automated bulk material intake mechanism.
Decision	The operator is required to manually authorize the transition

Constraint	between each manufacturing stage. This is executed exclusively through the touchscreen GUI.
Notice & Alert Constraint	The system is limited to utilizing audio-visual cues only to prompt the user to either proceed to the next automated sequence or revert to a previous state.

Table 13: Human-Machine Operational Constraints

References:

[1] Alabi, O. O., Adeyi, T., & Ekun, S. (2023). *Analyzing energy performance and assessing dry moisture content of briquettes through numerical investigations*. Engineering and Technology Journal, 42(1), 173–182. https://doi.org/10.30684/etj.2023.143782.1608

[2] Alfanz et al. (2023). *Smart farm agriculture design by applying a solar power plant*. https://doi.org/10.25077/jnte.v12n2.1085.2023

[3] Alora, L. B., & Amaya, F. L. (2025). *Impact of lean principles on the productivity of bottling line process at XYZ manufacturing company*. International Journal of Academic Research in Business and Social Sciences, 15(4). https://doi.org/10.6007/IJARBSS/v15-i4/25188

[4] Anonas, S. D. S., Eugenio, F. D. T., Flores, B.-H. F., Balite, P. H. M., Tomacruz, J. G. T., Limjuco, L. A., & Ocon, J. D. (2023). *From waste to renewable energy: A policy review on waste-to-energy in the Philippines*. *Sustainability, 15*(17), 12963. https://doi.org/10.3390/su151712963

[5] Arabi, & Dahmardeh Ghalehno. (2022). *Suitability of thin-layer drying models for halogen lamp drying of sugarcane bagasse*. https://doi.org/10.15376/biores.18.1.1052-1071

[6] Baccay, O. T. (2026). *DENR turns paper waste into eco-friendly charcoal briquettes in Batanes*. Philippine Information Agency. https://pia.gov.ph/news/denr-turns-paper-waste-into-eco-friendly-charcoal-briquettes-in-batanes/

[7] Barkalov, A., Titarenko, L., Bieganowski, J., & Krzywicki, K. (2024). *Basic approaches for reducing power consumption in finite state machine circuits-A review*. *Applied Sciences, 14*(7), 2693. https://doi.org/10.3390/app14072693

[8] Burigsay, H. L. (2025). *Developing sustainable solid waste management enterprise in Pudtol, Apayao: A feasibility study and program design*. *International Journal for Multidisciplinary Research, 7*(3). https://www.ijfmr.com/research-paper.php?id=44958

[9] Cabaces, D., et al. (2023). *Development of a low-cost briquetting system utilizing agri-waste materials as a renewable energy fuel*. Batangas State University. https://ojs.batstate-u.edu.ph/index.php/IRJIEST/article/view/94

[10] Castillo-Dida, V. (2024). *The link between electricity prices and carbon emissions*.https://journals.uplb.edu.ph/index.php/EDJ/article/view/1266

[11] Carrión-León, D. I., Lopez-Ramos, M. P., Santillan-Valdiviezo, L. G., Tanguila-Tapuy, D. S., Morocho-Santos, G. M., Moyano-Arias, R. J., Yautibug-Apugllón, M. E., & Chacón-Luna, A. E. (2025). *Evaluating interaction capability in a serious game for children with ASD: An operability-based approach aligned with ISO/IEC 25010:2023*. *Computers, 14*(9), 370. https://doi.org/10.3390/computers14090370

[12] Chipangura et al. (2024). *Fabrication of briquettes from charcoal fines using tannin formaldehyde resin as a binder*. https://doi.org/10.29333/ejosdr/14125

[13] Drutchas, J., & Eppinger, S. (2022). *Guidance on application of agile in combined hardware and software development projects*. *Proceedings of the Design Society, 2*, 151-160. https://doi.org/10.1017/pds.2022.16

[14] Dumagan, J. C., & Abrigo, M. R. M. (2022). *Rational choices and welfare changes in Philippine family energy demand*. De La Salle University.

[15] Dumrique, M.-C. G., Capili, J. M. F., Macugay, J. T., Pasilabban, E. T., & Alvarez, J. P. (2024). *Production and evaluation of carbonized rice straw and corn husk briquettes as a sustainable and alternative fuel for cooking*. *Journal of Pure and Applied Sciences, 4*(2). https://csu.org.ph/jpas/article/download/54/47/191

[16] Escalante, & Alcaraz. (2025). *Performance of a locally developed automated screw-type charcoal briquetting machine*. https://scienggj.org/2025/SciEnggJ%202025-vol18-no01-p44-50-Escalante%20and%20Alcaraz.pdf

[17] Fagcang, H., Stobart, R., & Steffen, T. (2022). *A review of component-in-the-loop: Cyber-physical experiments for rapid system development and integration*. *Advances in Mechanical Engineering, 14*(8). https://doi.org/10.1177/16878132221109969

[18] Felongco, W. S., Hemongala, E. E. M., Remotin, P. I. L., & Villacarlos, R. F. (2025). *Innovative conversion of paper wastes into eco-friendly charcoal briquettes as a sustainable energy source and solution for school-based waste management*. Ignatian International Journal for Multidisciplinary Research, 3(11), 827–841. https://doi.org/10.5281/zenodo.17762647

[19] Figueiroa, V., & Torres, J. P. N. (2022). *Simulation of a small smart greenhouse*. *Designs, 6*(6), 106. https://doi.org/10.3390/designs6060106

[20] Genuino, A. F. R., Tancuan, J. M. M., et al. (2022). *Evaluation of bio briquettes made from Musa acuminata Colla, Musa acuminata and Musa balbisiana Silk*. De La Salle University.

[21] Glalah et al. (2024). *Binder-type effect on physico-mechanical, combustion and emission properties of biochar briquettes*. https://doi.org/10.1371/journal.pone.0306827

[22] Gupta, S. S., Prasad, R., Chakraborty, D., & Ebbing, M. (2026). Pelletizing poultry litter with biochar: A strategy to enhance poultry litter utilization. Waste and Biomass Valorization. https://www.researchgate.net/publication/402101786_Pelletizing_Poultry_Litter_with_Biochar_A_Strategy_to_Enhance_Poultry_Litter_Utilization

[23] Habib et al. (2023). *Estimating shifts in fuel stacking among solid biomass fuels and LPG*. https://doi.org/10.21203/rs.3.rs-2674609/v1

[24] Ibitoye, A. R., et al. (2022). *Combustion, physical, and mechanical characterization of composites fuel briquettes*. International Journal of Renewable Energy Development, 11(3), 859-868. https://ijred.cbiore.id/index.php/ijred/article/view/41290

[25] *International Organization for Standardization & International Electrotechnical Commission*. (2023). *ISO/IEC 25010:2023 Systems and software engineering - Systems and software Quality Requirements and Evaluation (SQuaRE) - Product quality model*. https://www.iso.org/standard/78176.html

[26] Jesus, C., Amorim, I., Pontes, E., Gonçalves, S., Moradbeikie, A., & Lopes, S. (2025). *Impact of adopting industry 4.0 practices and technologies on time and*

motion study: a pilot study revealing efficiency gains in a multi-operation workstation. Production & Manufacturing Research, 13(1). https://www.tandfonline.com/doi/full/10.1080/21693277.2025.2476436

[27] Kalli et al. (2022). *Subsidized LPG scheme and the shift to cleaner household energy use*. https://doi.org/10.3390/su14042450

[28] Lomunyak et al. (2024). *Characterization, optimization and emission analysis of manually-made charcoal dust briquettes with starch, paper and algae binders*. https://doi.org/10.1016/j.heliyon.2024.e40991

[29] Łaska, & Ige. (2023). *A review: Assessment of domestic solid fuel sources in Nigeria*. https://doi.org/10.3390/en16124722

[30] Magbanua, S. A. (2026). *LPG use drops by 30% as Filipinos shift to firewood in summer*.https://tribune.net.ph/2026/04/13/lpg-use-drops-30-filipinos-shift-to-firewood-in-summer

[31] Malovanyy et al. (2023). *The use of binders of natural origin to improve the technology of creating fuel briquettes from wood waste*. https://doi.org/10.12911/22998993/171901

[32] Meniano, S. (2026). *DENR-8 pushes eco-friendly charcoal as LPG prices drive poaching*. Philippine News Agency.https://www.pna.gov.ph/articles/1273864

[33] Mentes, D., Jordan, A., Farkas, L., Muranszky, G., Fiser, B., Viskolcz, B., & Poliska, C. (2024). *Evaluating emissions and air quality implications of residential waste incineration*. *Scientific Reports, 14*, 21314. https://doi.org/10.1038/s41598-024-72173-w

[34] Mofokeng, N. N., Madikizela, L. M., Tiggelman, I., & Chimuka, L. (2024). *Chemical profiling of paper recycling grades using GC-MS and LC-MS: An exploration of contaminants and their possible sources*. *Waste Management, 189*, 148-158. https://doi.org/10.1016/j.wasman.2024.08.014

[35] Montiel et al. (2023). *Algal biomass (Green algae) as a binding agent for the production of eco-briquettes from water hyacinth*. De La Salle University.

[36] Mwamlima et al. (2023). *Efficiency of place-based innovated briquette making technologies for sustainable cooking energy*. https://doi.org/10.37425/eajsti.v4i3.710

[37] Nkuriyingoma, O., Özdemir, E., & Sezen, S. (2022). *Techno-economic analysis of a PV system with a battery energy storage system for small households: A case study in Rwanda*. Frontiers in Energy Research, 10. https://doi.org/10.3389/fenrg.2022.957564

[38] Obi, O. F., Pecenka, R., & Clifford, M. J. (2022). *A review of biomass briquette binders and quality parameters*. In Energies (Vol. 15, p. 2426). https://doi.org/10.3390/en15072426

[39] Pantuhan, G. P., Servinas, M. O., & Hassan, V. A. (2024). *Energy challenges and strategies by rural communities in a southern Philippine city*. *Journal of Health Research and Society, 3*(2), 66-77. https://doi.org/10.34002/jhrs.v3i2.59

[40] Pniel, M. V. H. (2026). Automation of moisture level measurement in charcoal briquettes. *E3S Web of Conferences*. https://doi.org/10.1051/e3sconf/202668701012

[41] Qi et al. (2022). *Improving combustion technology for cooking activities for pollutant emission reduction and carbon neutrality*. https://doi.org/10.3390/atmos13040561

[42] Sjöstrand, B., & Brolinson, A. (2022). *Addition of polyvinylamine in chemi-thermomechanical pulp and kraft pulp and the effects on dewatering, strength, and air permeance*. BioResources, 17(3), 4098–4115. https://bioresources.cnr.ncsu.edu/resources/addition-of-polyvinylamine-in-chemi-thermomechanical-pulp-and-kraft-pulp-and-the-effects-on-dewatering-strength-and-air-permeance/

[43] Tesfaye, A., et al. (2023). *Development and evaluation of a coffee husk biomass briquette machine*. Indian Journal of Sustainable Research. https://www.researchgate.net/publication/373444645_Development_and_Evaluation_of_a_Coffee_Husk_Biomass_Briquette_Machine

[44] Thongsan, T., Angkawisittpan, N., Thung-Od, K., & Techasarntikul, N. (2023). *Smart data management for charcoal briquettes manufacturing using IoT technologies*. Materials Today: Proceedings. https://doi.org/10.64289/iej.25.03A11.9182674

[45] Tuan et al. (2023). *Study on drying of bitter gourd slices based on halogen dryer*. https://doi.org/10.17221/97/2022-rae

[46] Waheed, A., et al. (2022). *An overview of torrefied bioresource briquettes: quality-influencing parameters, enhancement through torrefaction and applications*. Bioresources and Bioprocessing, 9, 23. https://link.springer.com/article/10.1186/s40643-022-00608-1

[47] Waluyo et al. (2022). *An effect of blending ratio on mechanical and thermal properties of sawdust-cocoa pod briquettes*. https://doi.org/10.18280/ijdne.170417

[48] wondimagegn, G. H., & Edao, S. A. (2023). *Adaptation and Evaluation of Pellet Press Briquette Machine*. https://iijsr.com/data/uploads/7481.pdf

[49] Yusuf et al. (2023). Characteristics of charcoal briquettes from rice husk waste with compaction pressure variations. https://doi.org/10.12911/22998993/159966